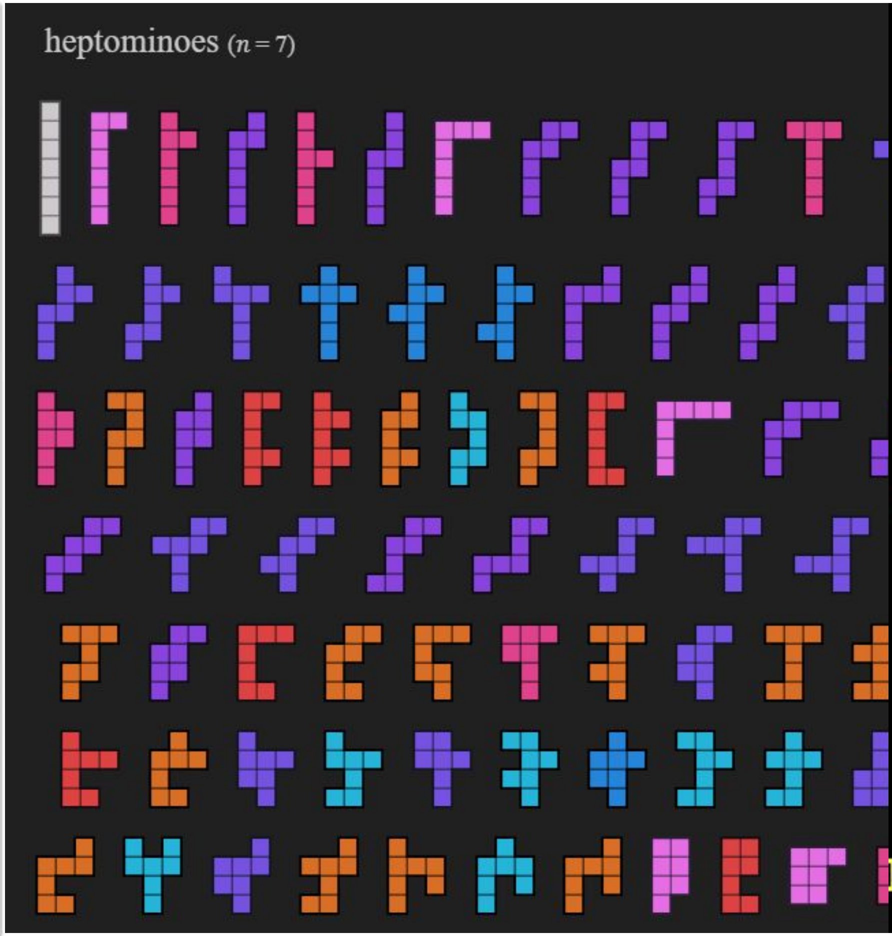
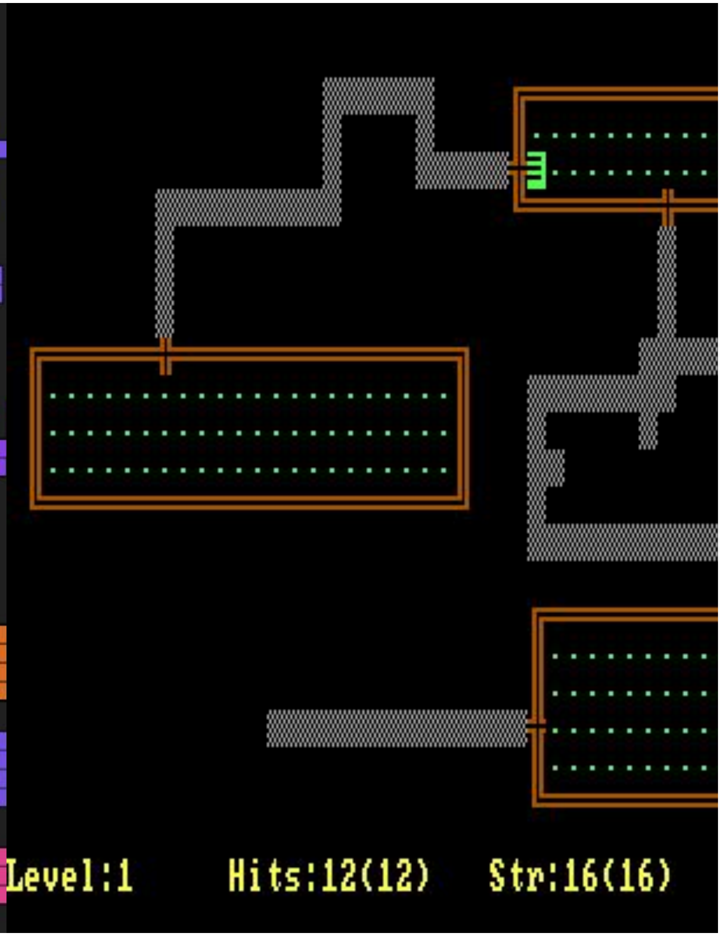




Sol LeWitt, Combinatorial Enumeration, and Rogue

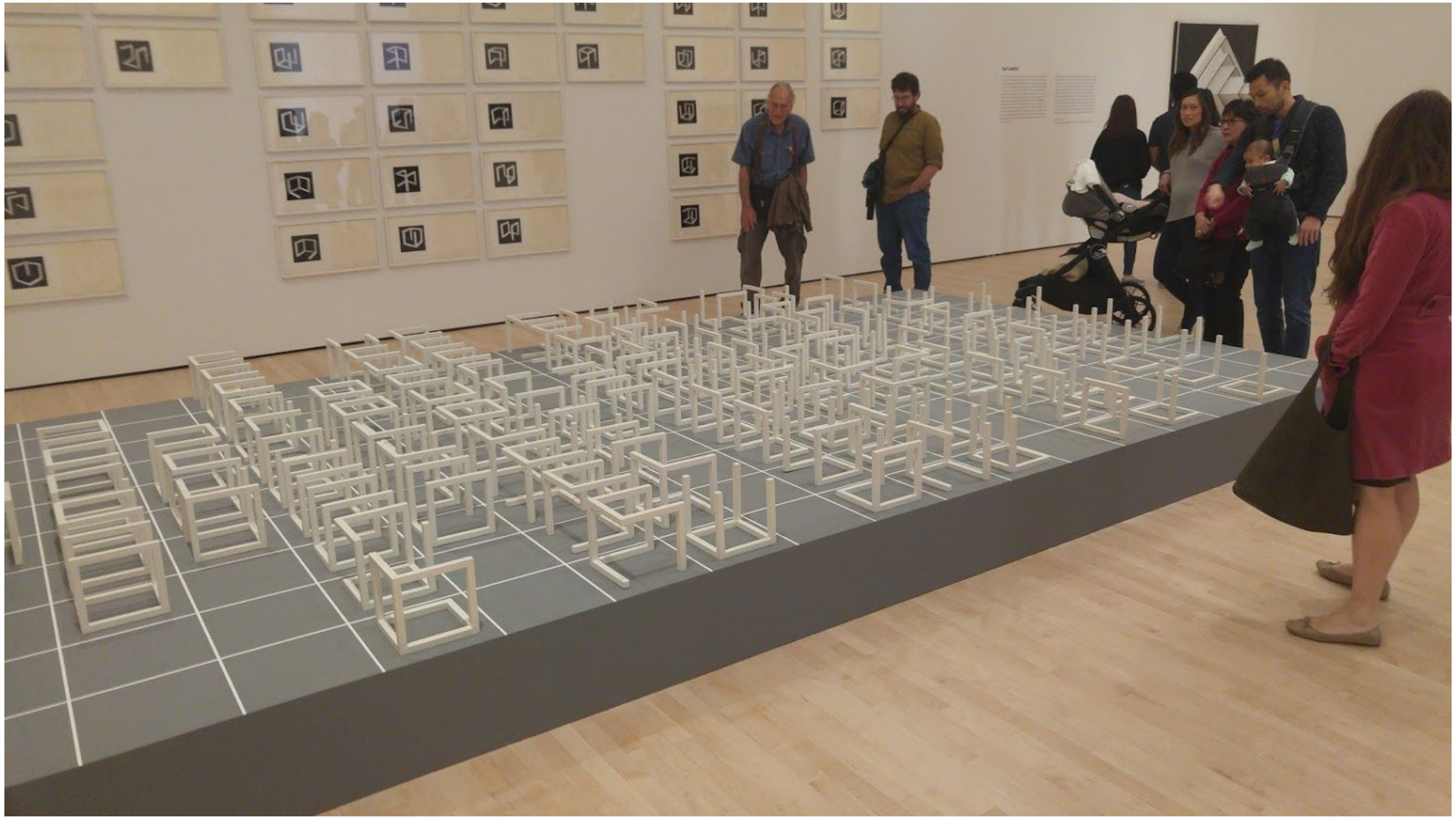
Mark Gritter
Minnebar 2026

Sol LeWitt, "'All One, Two, & Three-Part Combinations of Three Transparent Colors', 1985

Labyrinth of Polyominoes, heptominoes

Rogue (PC port)

Sol LeWitt



20th Century Conceptual Artist, 1928-2007

- wall drawings
- minimalism
- lots of cubes!

← “Incomplete Open Cubes”, 1974

Photo by the speaker, at San Francisco Museum of Modern Art, 2018

Wall drawings

Sol LeWitt, "Wall Drawing 289", 1976

A 6-inch (15 cm) grid covering each of the four black walls. White lines to points on the grids. Fourth wall: twenty-four lines from the center, twelve lines from the midpoint of each of the sides, twelve lines from each corner.

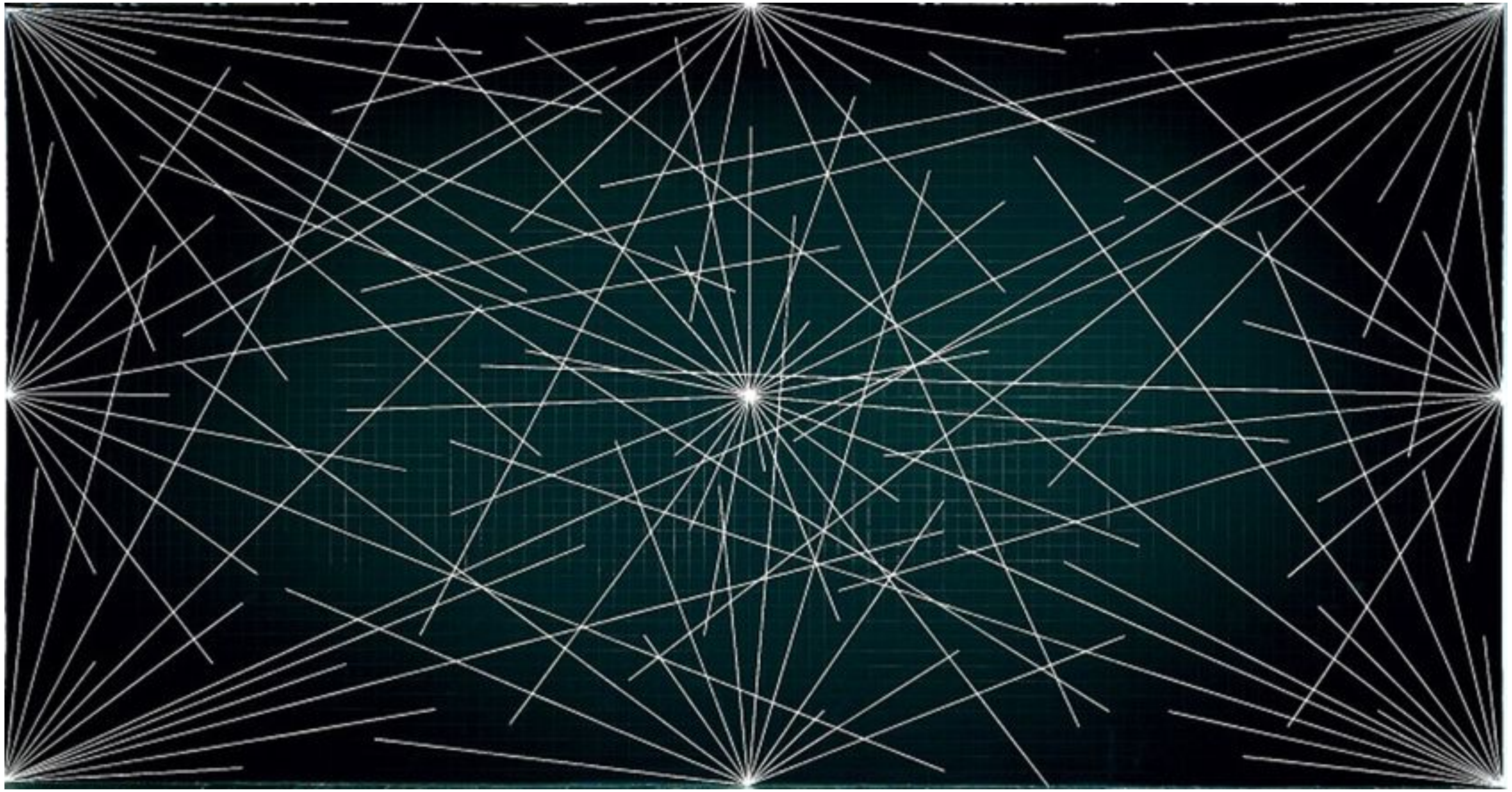


Image from https://whitney.org/collection/works/25530

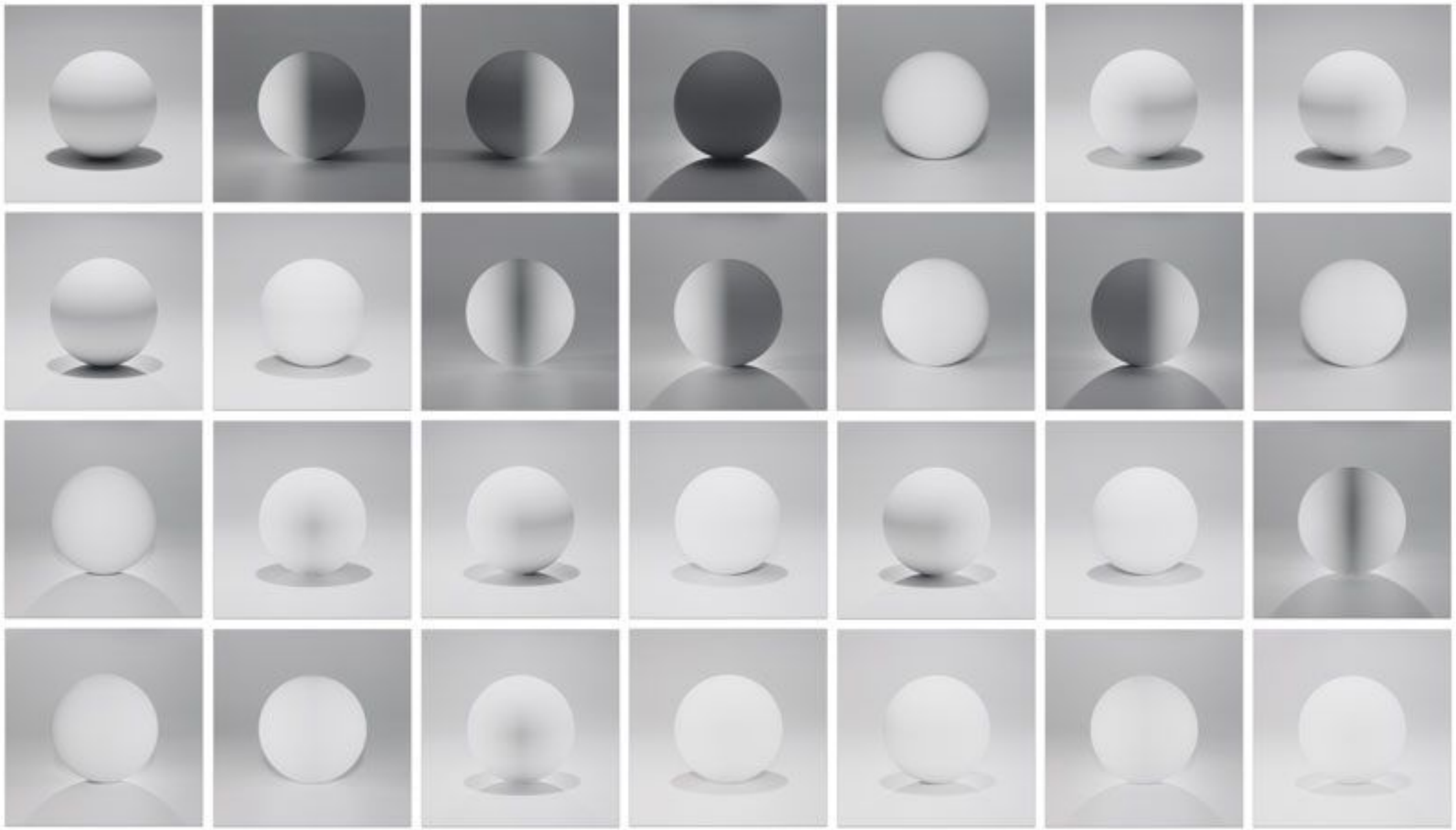
Wall drawings (2)

Wall drawings are generative art! They are given as **instructions** rather than as an image. (*)

"The only permanent, concrete form of Wall Drawing #289 is a set of typewritten guidelines and a certificate of authenticity signed by the artist… The exact angle and length of the lines are determined by those who draw them, and the work's precise configuration and scale may be adapted to fit a variety of architectural contexts. Consequently, the wall drawing can differ significantly with each realization." – Whitney Museum of American Art, https://whitney.org/collection/works/25530

(*) this separation is ruined a bit by having to hire a draftsperson from the LeWitt Studio to execute an "official" drawing.

Serial Art



Sol LeWitt, “A sphere lit from the top, four sides, and all their combinations”, 2004

Image from sollewittprints.org

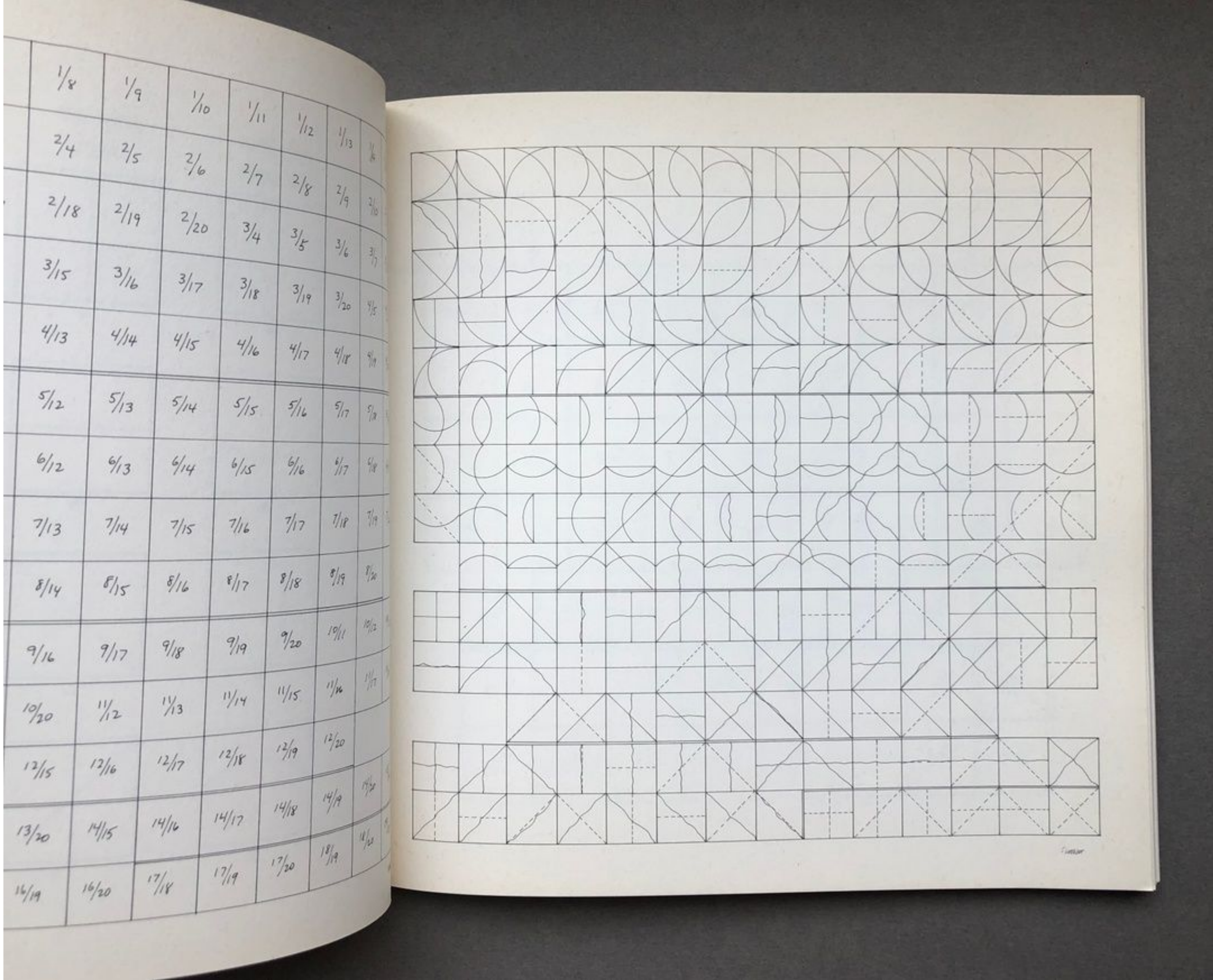
(P.S. there should be 32, or 31, not 28?)

“Arcs and Lines”, Sol LeWitt, 1974

“All combinations of arcs from four corners, arcs from four sides, straight lines, not-straight lines, and broken lines”

Image from https://www.printedmatter.org/catalog/14771/

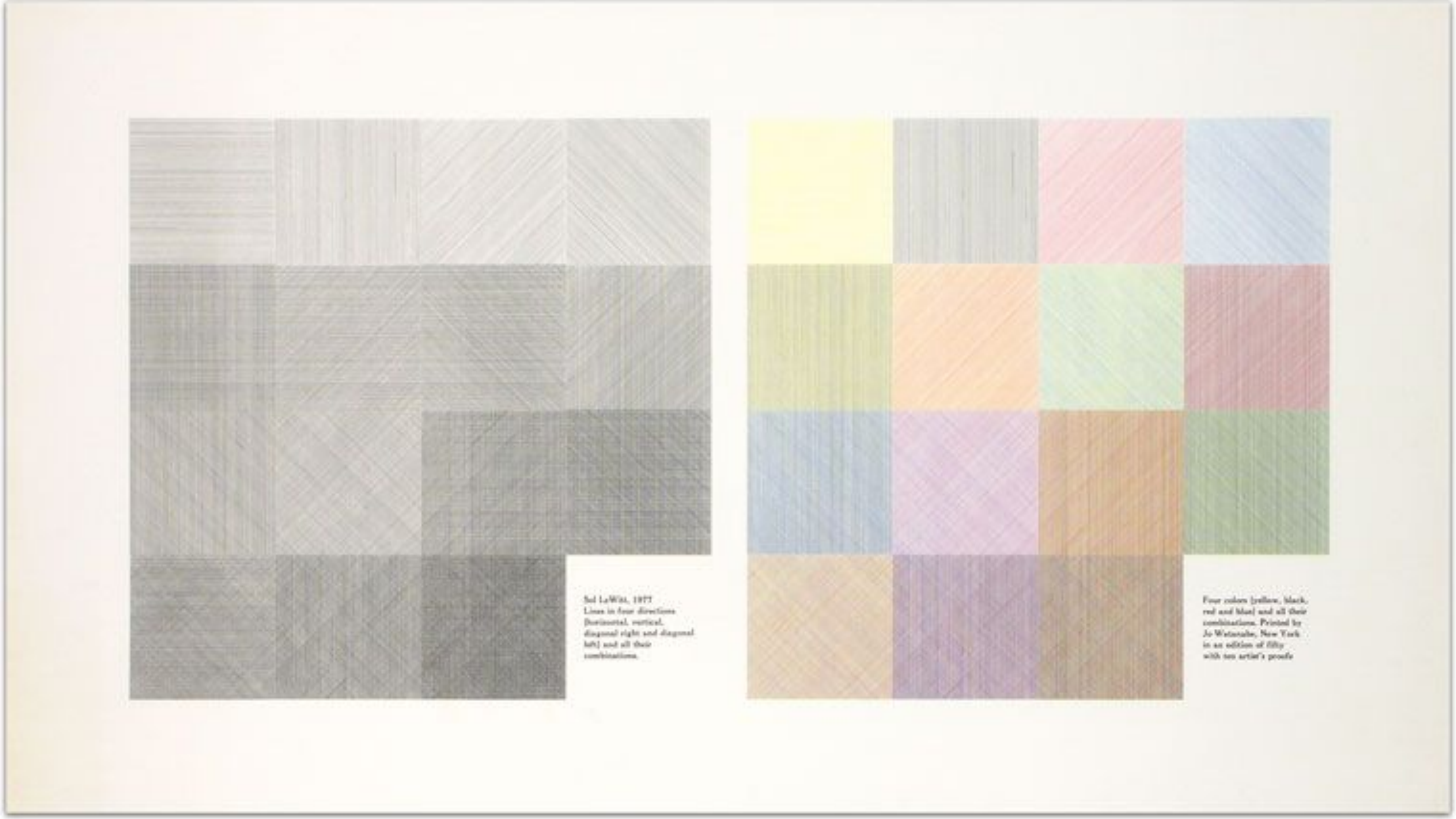
(A used copy will set you back about $200-$350 so I don’t, alas, own this.)



“Lines in Four Directions (Horizontal, Vertical, Diagonal Right and Diagonal Left) and All Their Combinations. Four Colors (Yellow, Black, Red and Blue) and All Their Combinations”

Sol LeWitt, 1978

Image from: https://www.sollewittprints.org/artwork/lewitt-raisonne-1978-01/



Serial Art (2)

This takes a further step backwards into abstraction. The artwork is a description of a condition or axiom or law, which is then carried out to produce a tangible result. The artwork typically juxtaposes all of these results together as a finished work.

"*The serial artist does not attempt to produce a beautiful or mysterious object but functions merely as a clerk cataloguing the results of his premise.*" – Sol LeWitt

(I disagree, I think these *are* beautiful and mysterious. – Mark)

Serial Art (3)

"Three basic operating assumptions separate serially ordered works from multiple variants:

1. The derivation of the terms or interior divisions of the work is by means of a numerical or otherwise systematically predetermined process (permutation, progression, rotation, reversal).

2. The order takes precedence over the execution.

3. The completed work is fundamentally parsimonious and systematically self exhausting."

– Mel Bochner, "The Serial Attitude", Artforum, December 1967

(Of course, not everybody agrees with this either – Mark)

Coplans' Critique

Coplans defined Serial Art within a context of thematic variation, while also underscoring the importance of repetitive elements or images within an artwork or series. – https://www.eykynmaclean.com/exhibitions/the-serial-attitude

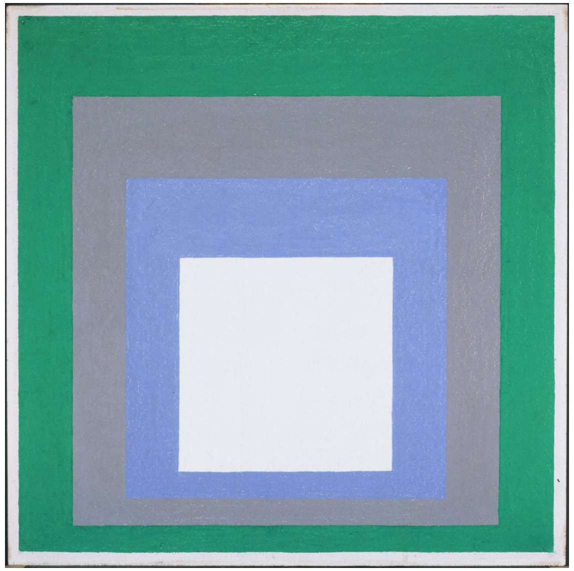
Though often painted in sets, that is, in a limited number that satisfy a given condition, serial images ***are nevertheless capable of infinite expansion****. There is no limit to the quantity of works in a series other than what is determined by the artist. Once established, a series may be kept open and added to periodically in the future…*

– John Coplans, "Serial Imagery", Artforum, October 1986

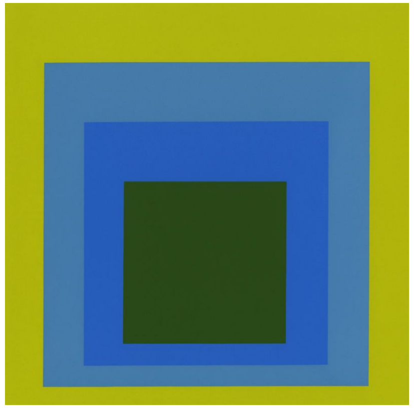
https://www.artforum.com/features/with-serial-imagery-the-idea-of-the-masterpiece-is-abandoned-211011/

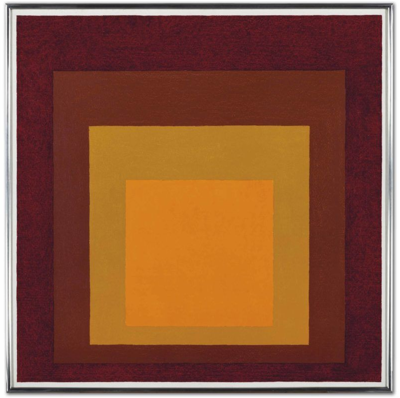
"[Bochner] evidently is unaware of the Dedekind-Cantor theory, which is central to the mathematical concept of serial forms." **(!!!)**

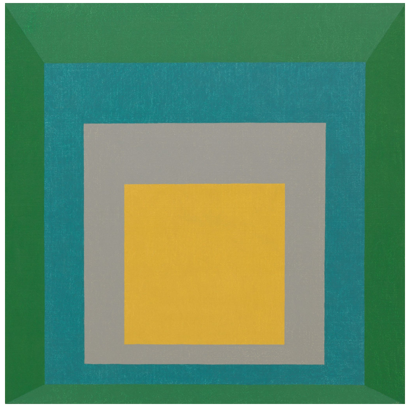
Josef Albers, “Homage to the Square”. Photo by Selena N. B. H. (cc-by-2.0) at the Tate Modern.



“a rigorously formulaic project comprising more than one hundred paintings and prints and developed over twenty-five years.” – MOMA.org





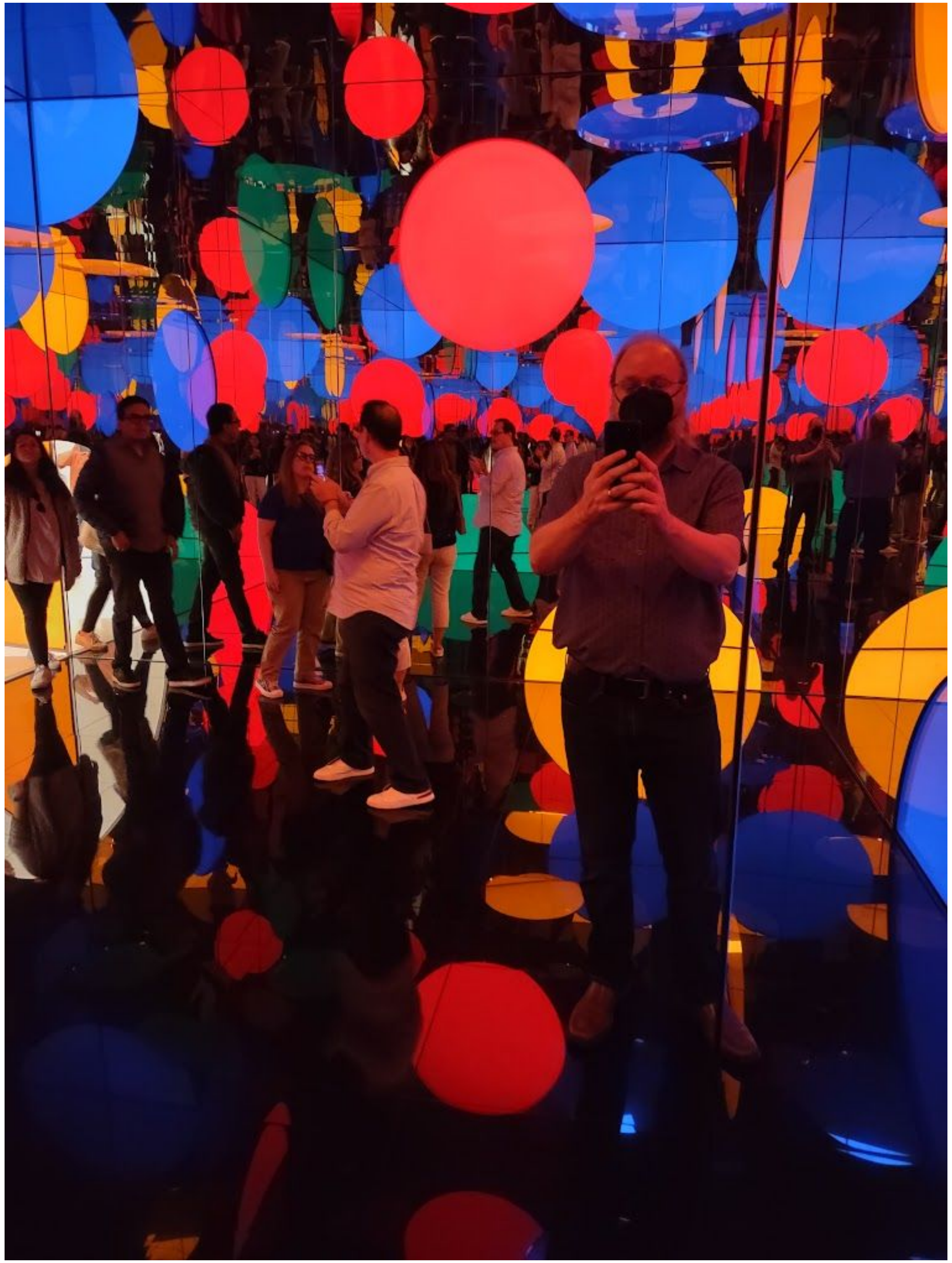


Yayoi Kusama, "Infinity Mirrored Room - Aftermath of Obliteration of Eternity", 2009

from https://hirshhorn.si.edu/kusama/infinity-rooms/#aftermath

"Passing Winter" (2005)
https://www.walkerart.org/collections/artwork/passing-winter/

"Dreaming of Earth's Sphericity, I Would Offer My Love (2023)" – September 2024 at SFMOMA (photo by speaker)



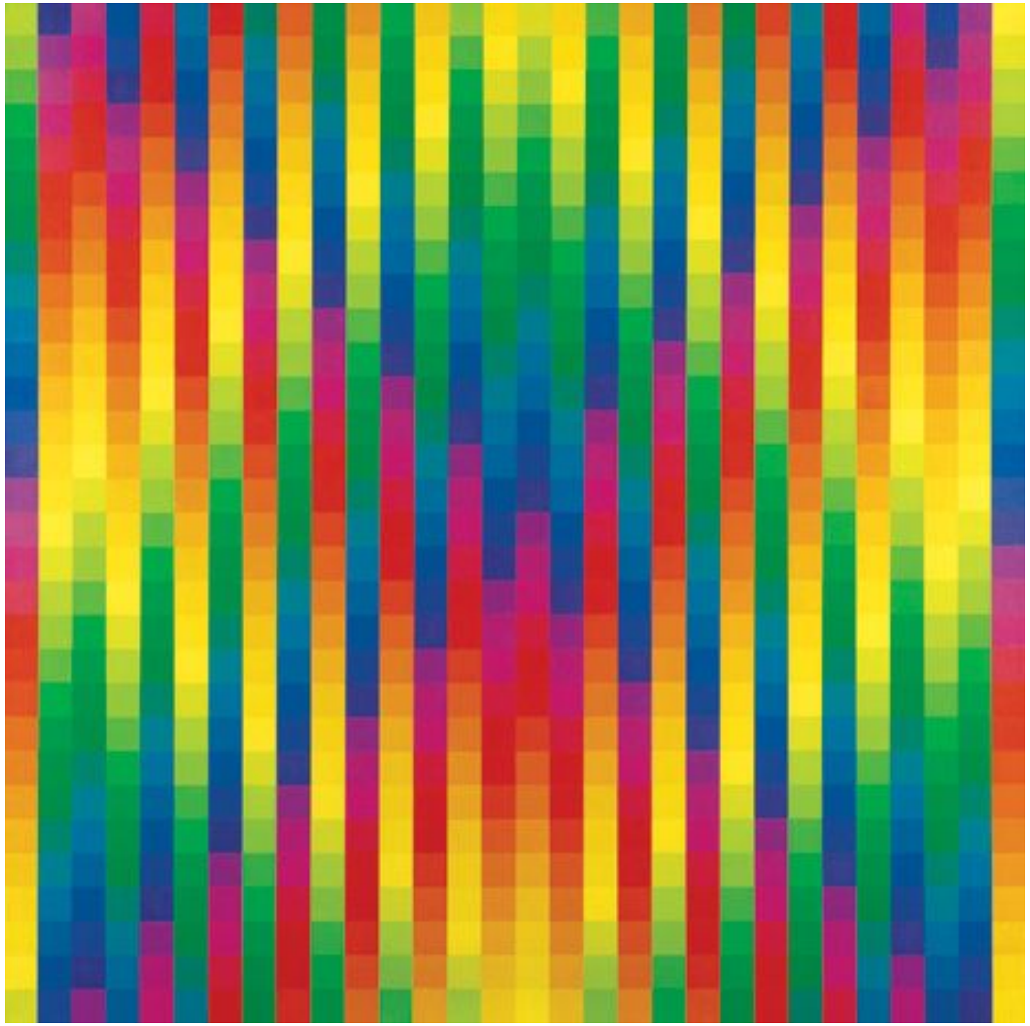
Hanne Darboven, (Süd-) Koreanischer Kalender, 1991

https://spruethmagers.com/artists/hanne-darboven/

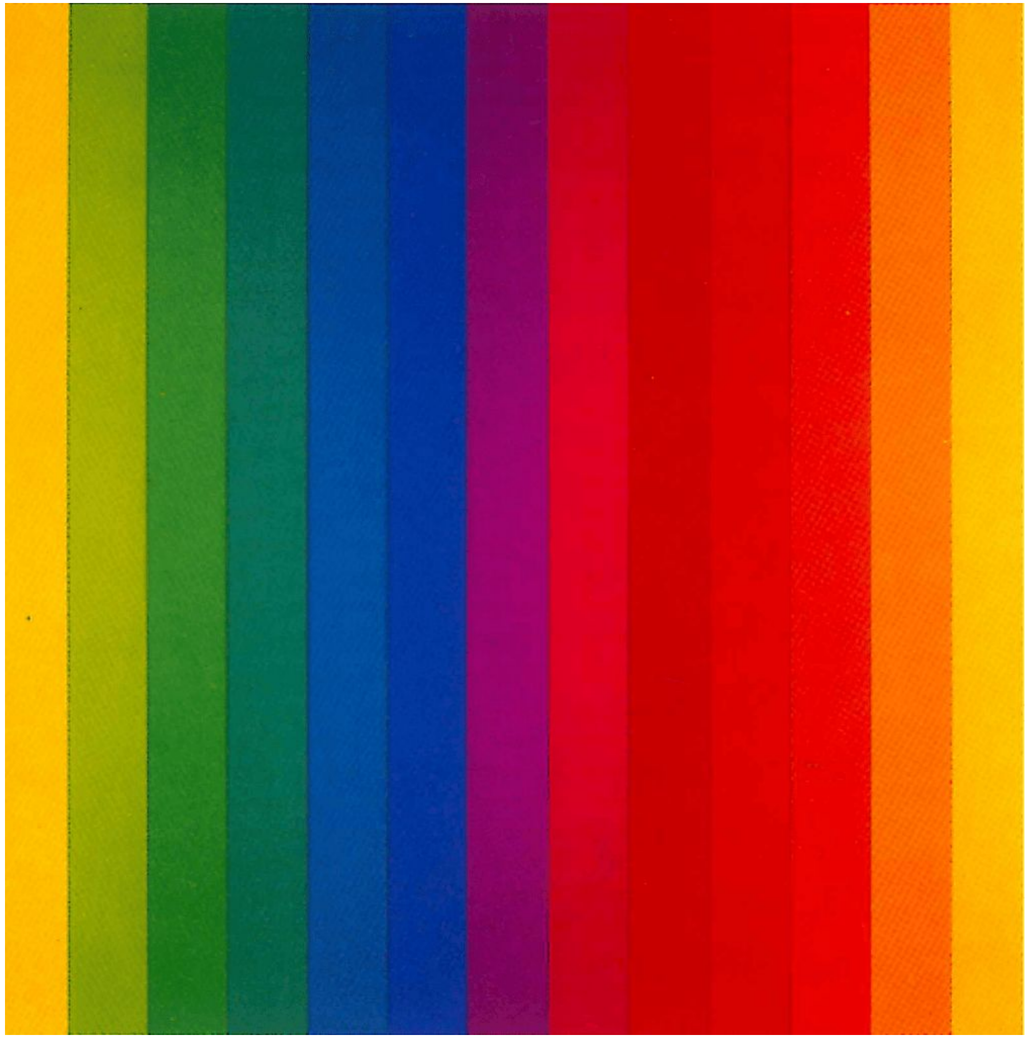
Mel Bochner, “36 Photographs and 12 Diagrams”, 1966, https://www.glenstone.org/artworks/36-photographs-and-12-diagrams

Mel Bochner, “Counting (Double Over)”, 1995, https://www.melbochnerprints.org/krakow-1994-01/#198

Richard Paul Lohse, “Dreissig vertikale systematische Farbreihen in gelber Rautenform”, 1943/1970
from https://www.lohse.ch/works_paintings3_e.html



Ellsworth Kelly, “Spectrum IV”, 1967
https://www.miandn.com/exhibitions/ellsworth-kelly/installation-views?view=thumbnails



Samuel Beckett's "Quad"

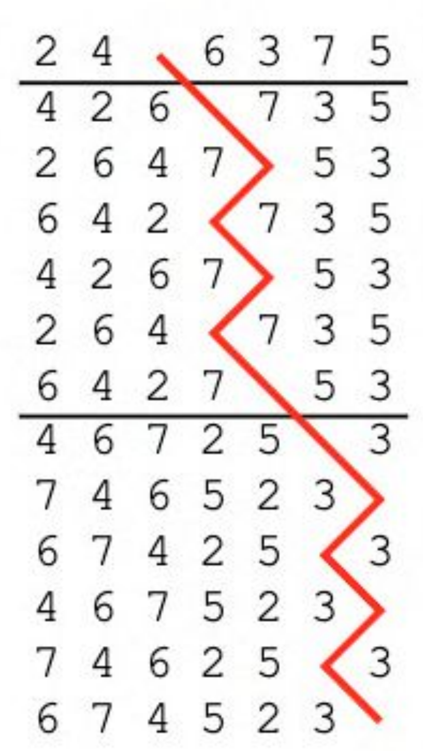
An experimental TV broadcast. Four performers move in and out of a square in a way that exhausts all combinations of presence and absence, one entering or leaving at a time:

https://en.wikipedia.org/wiki/Quad_(play)

But the exact parameters he wanted aren't possible for n=4! (He wanted the performer on stage the longest to be the one to leave.)

This led to the notion of "Beckett-Gray" codes, which do exist for other numbers: https://combinatorialpress.com/jcmcc-articles/volume-126/a-note-on-beckett-gray-codes-and-the-relationship-of-gray-codes-to-data-structures/

Other Serial Art

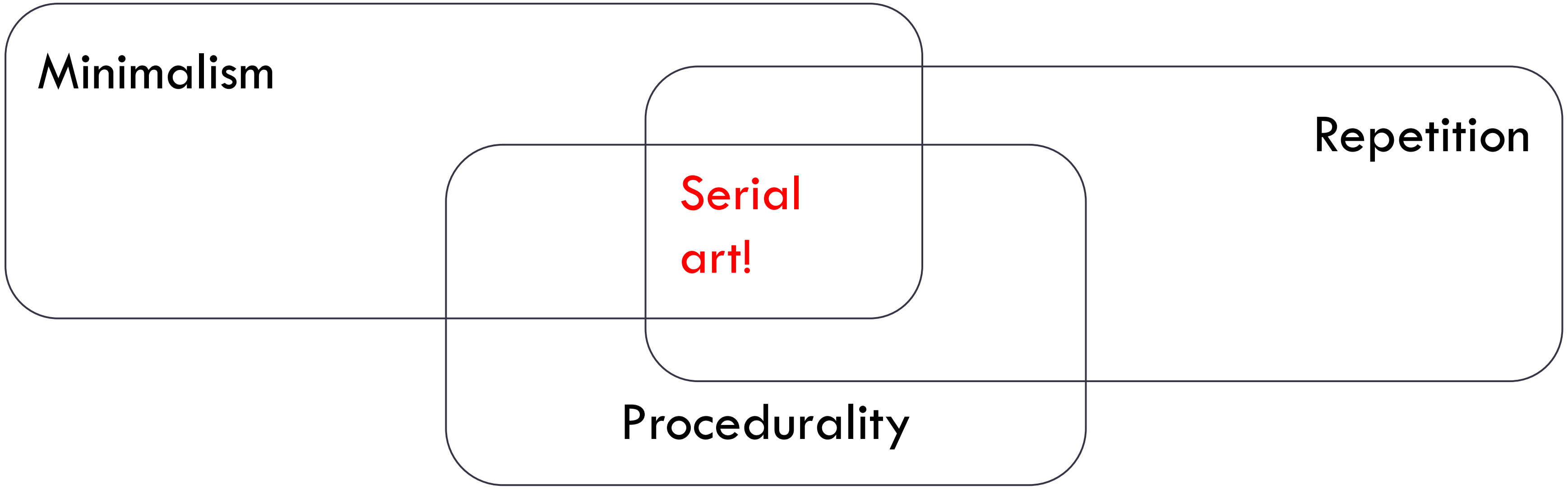


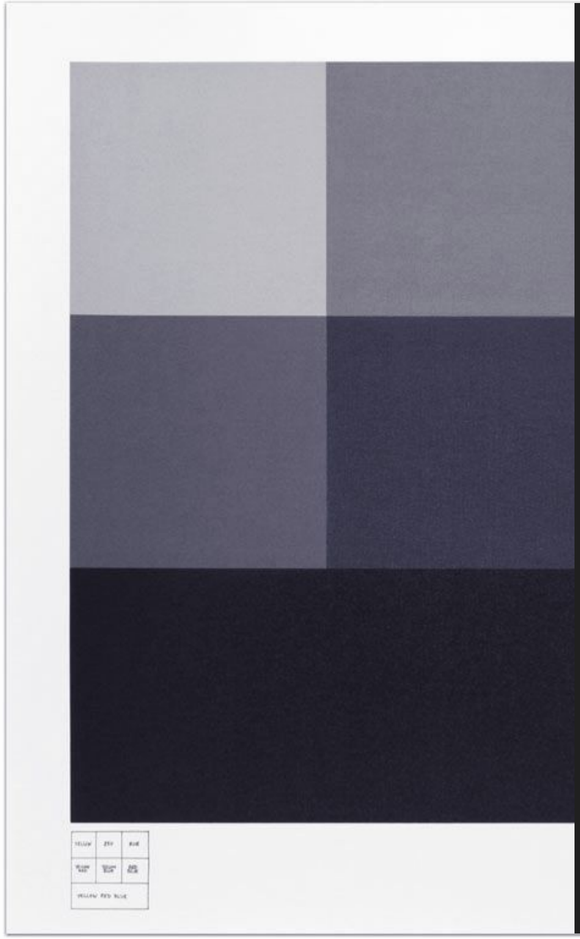
"Change Ringing" – playing permutations of a set of bells

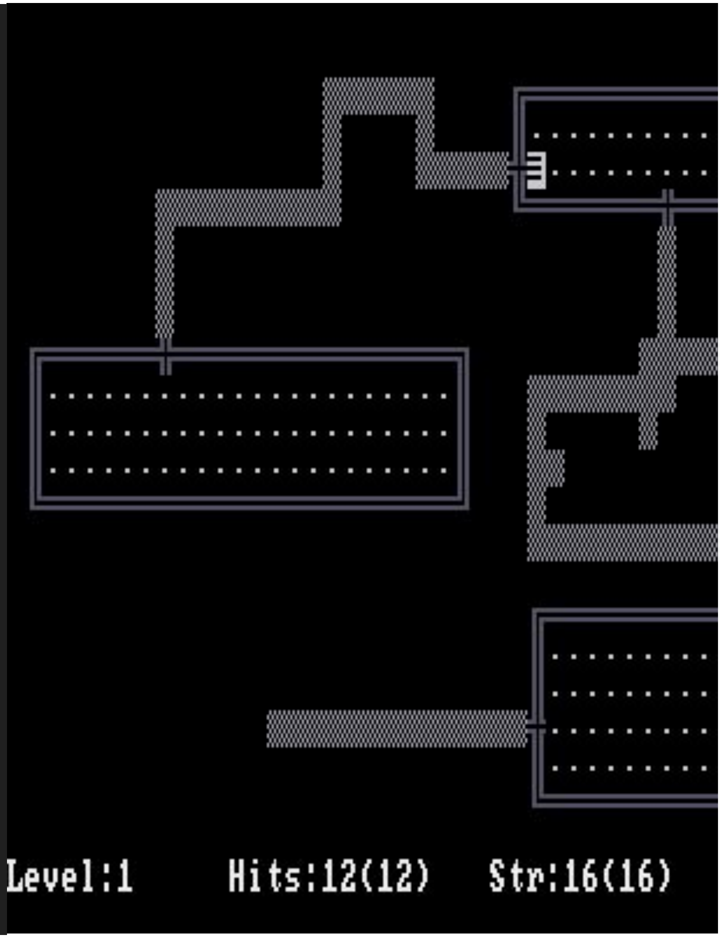
- Compositions are called "methods": each player is given a chart showing how their position in the sequence varies
- 7! orders = 5040 changes in a complete method.
- "Composers tend to base their methods in mathematical principles such as group theory" – https://harpers.org/archive/2025/10/a-change-of-tune-veronique-greenwood-bell-ringing/
  - You can tell it's contemporary art because there is active controversy about *what patterns are allowed*

The idea becomes a machine that makes the art.

– Sol LeWitt, “Paragraphs on Conceptual Art”, 1967







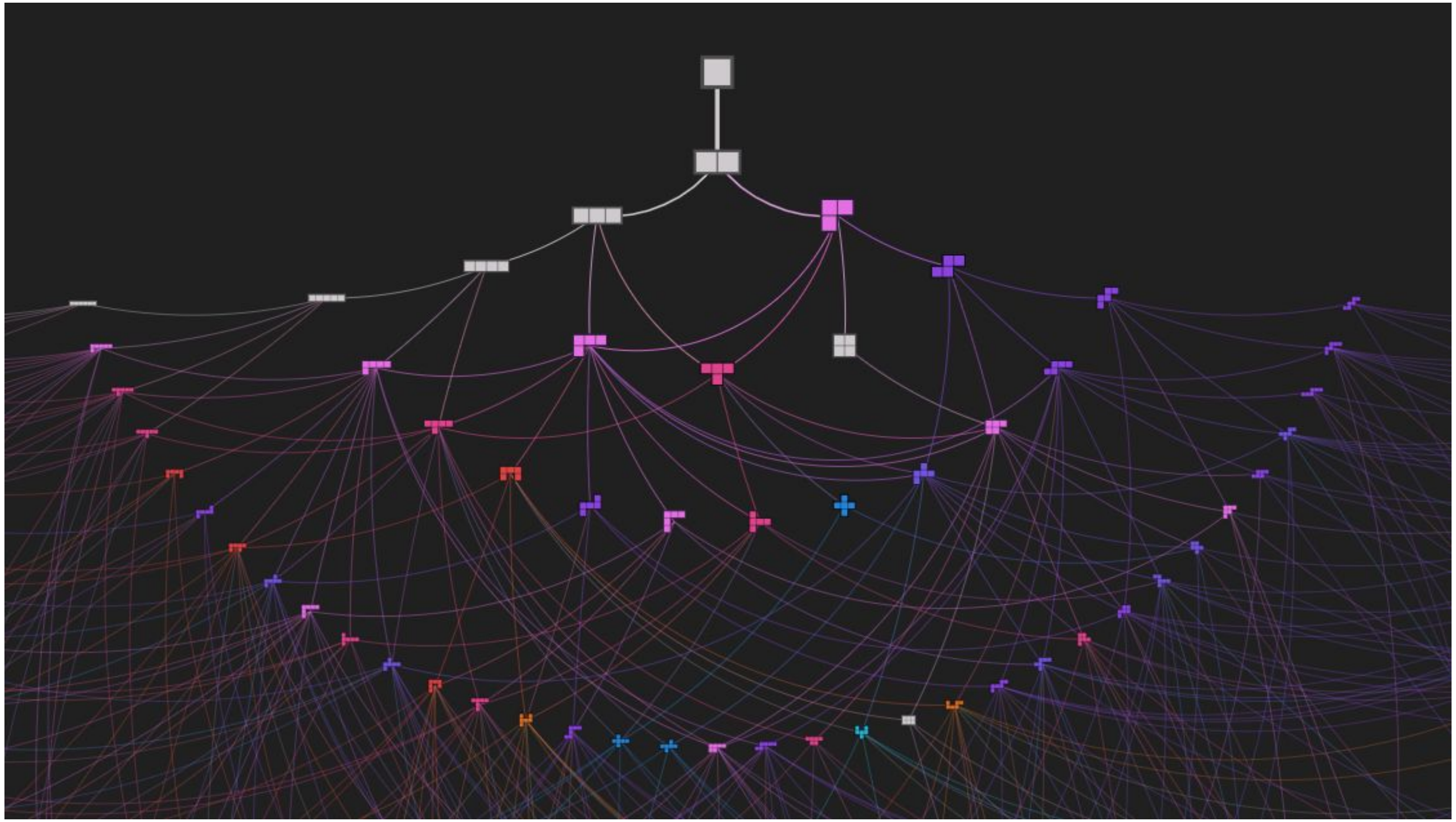
Enumerating All the Things

The Labyrinth of Polyominoes

By Roguelike Celebration 2024 speaker @tesseralis!

All polyominoes (up to n=8) here arranged by "genealogy".

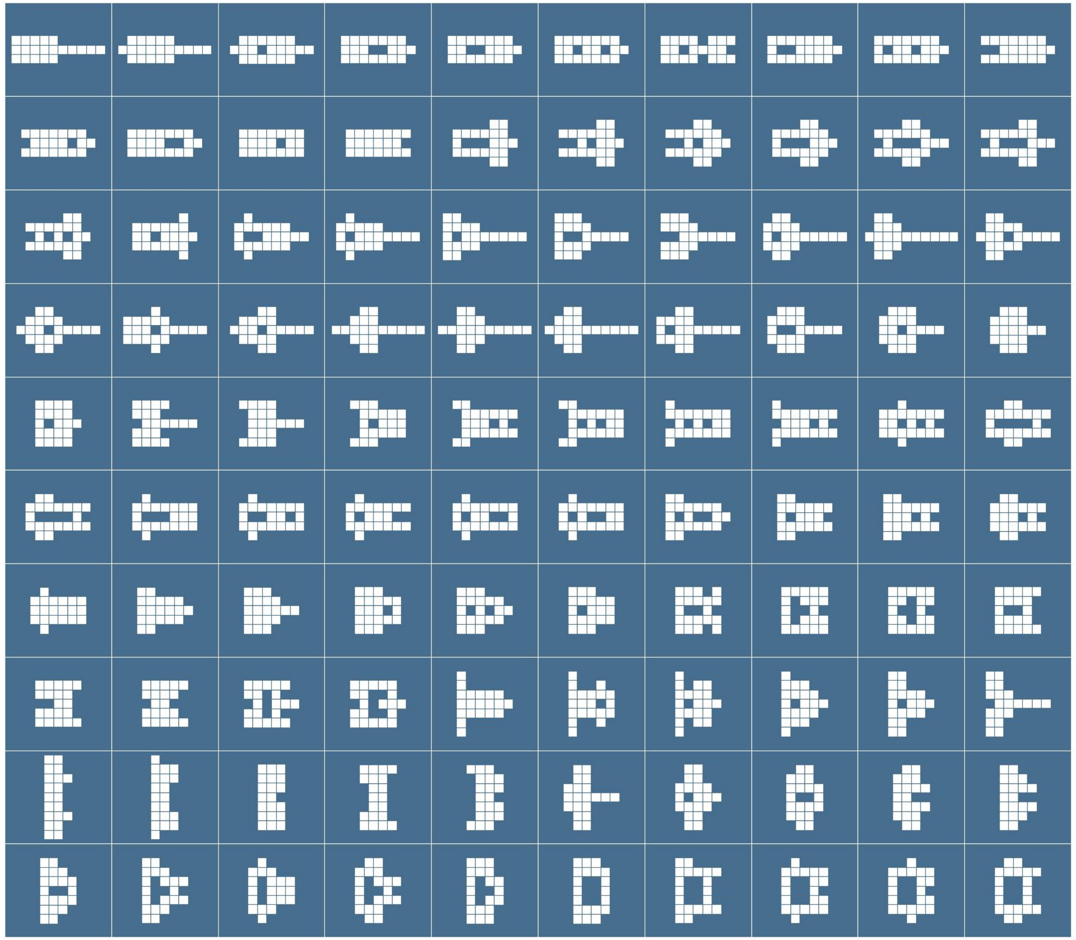
https://minos.tessera.li/



All symmetric shapes constructible from the tetrominoes

Jacob Siehler, 2025

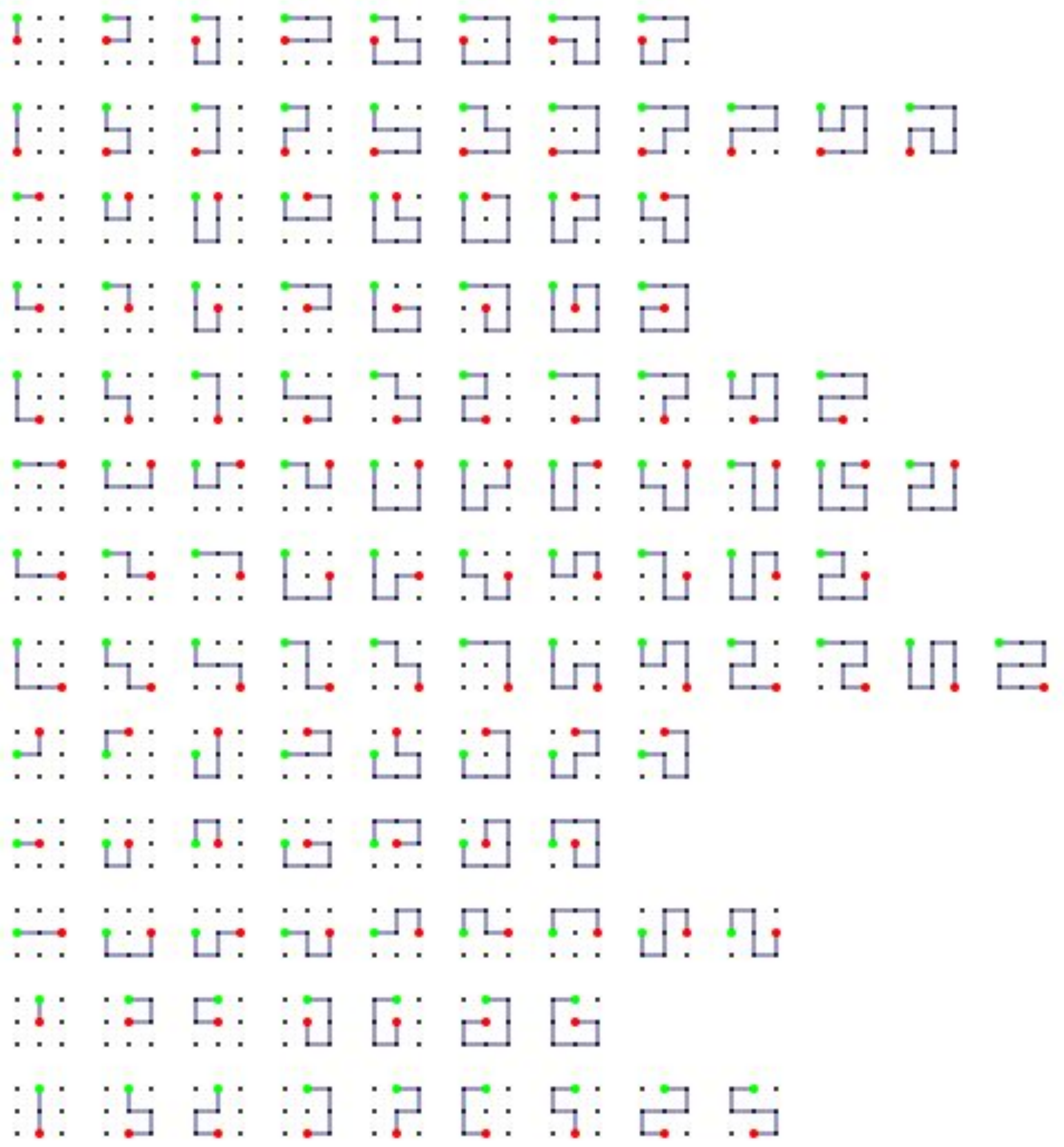
https://mathstodon.xyz/@jsiehler/114744887537037820



All simple paths on a 3x3 lattice

Visualization by the speaker

These show directed paths from the green point to the red point, up to symmetries of the rectangle (so, flips but not rotations) and interchange of start and end.

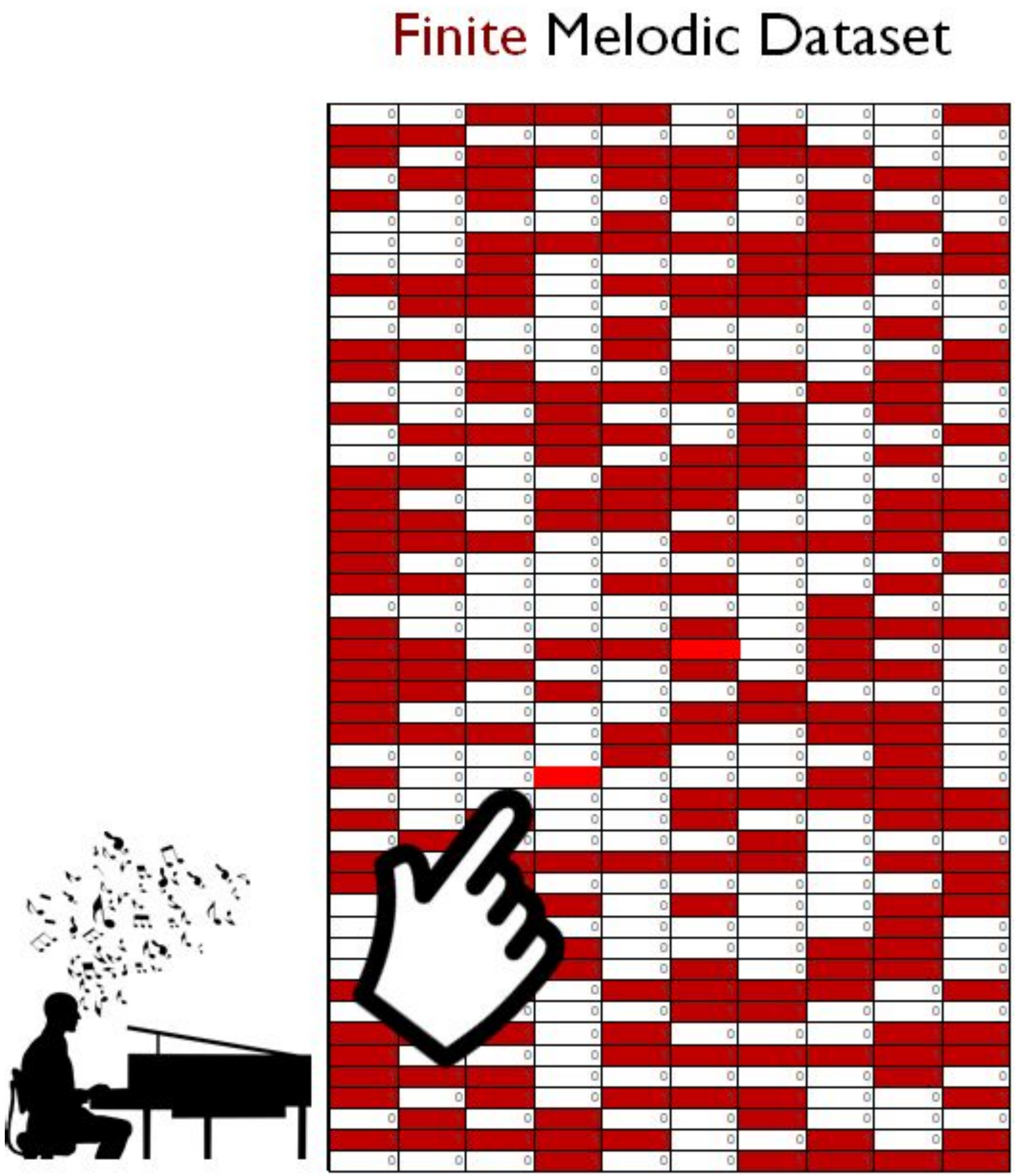


All the Music

Damien Riehl and Noah Rubin, https://allthemusic.info/

1. Generate every possible melody, up to 10 or 12 notes long.
2. Copyright them all.
3. Stop "substantially similar" music lawsuits forever?

Argument: either Riehl has copyrighted every unused melody, and they are free to use, or melodies aren't copyrightable.

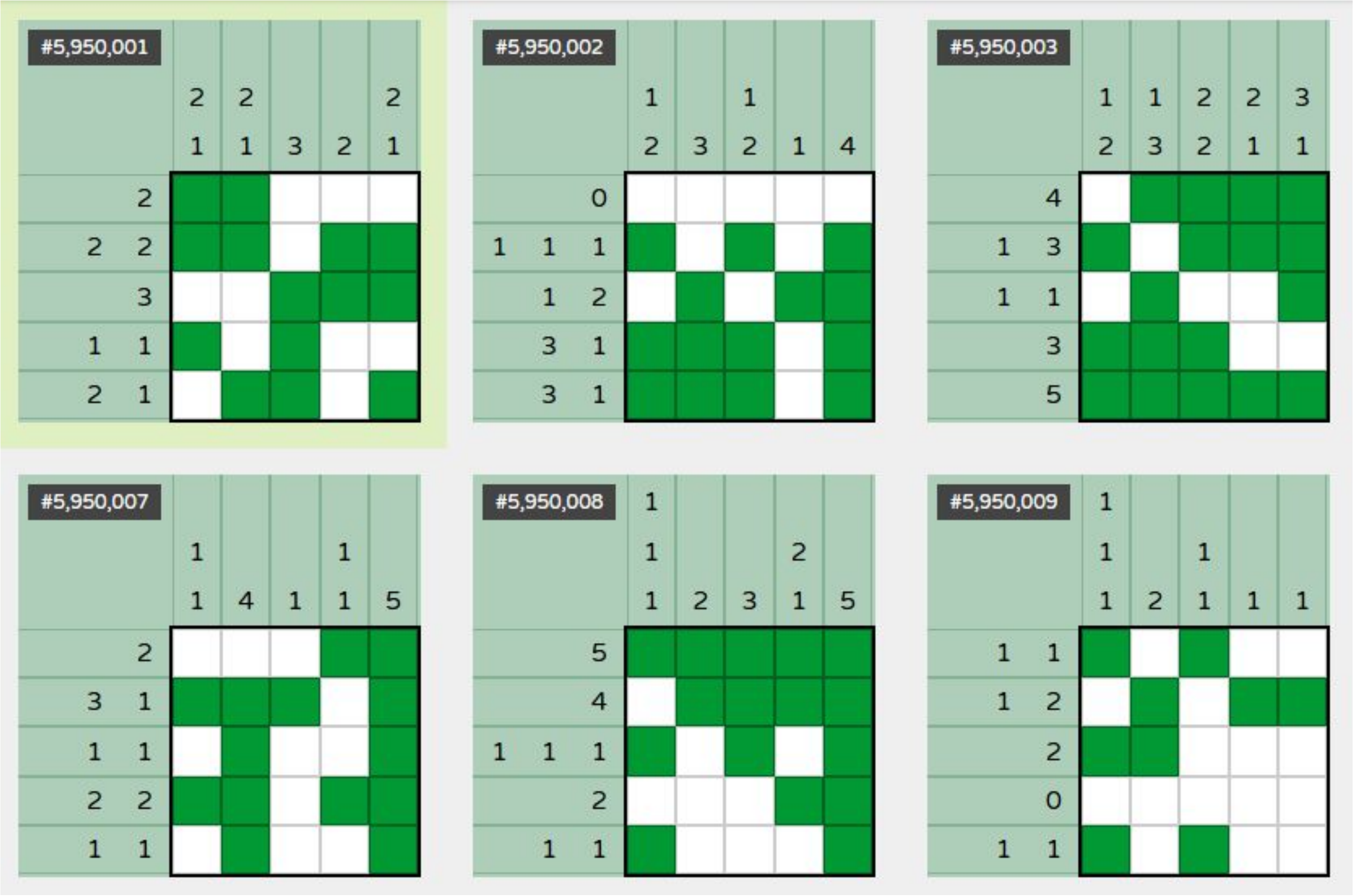


P.S. in the age of generative AI you should be able to find the unfortunate flaw in this argument.

Every 5x5 Nonogram

Okayest Studio, https://pixelogic.app/every-5x5-nonogram

24,976,511 possibilities, all solved by humans via a collaborative effort.

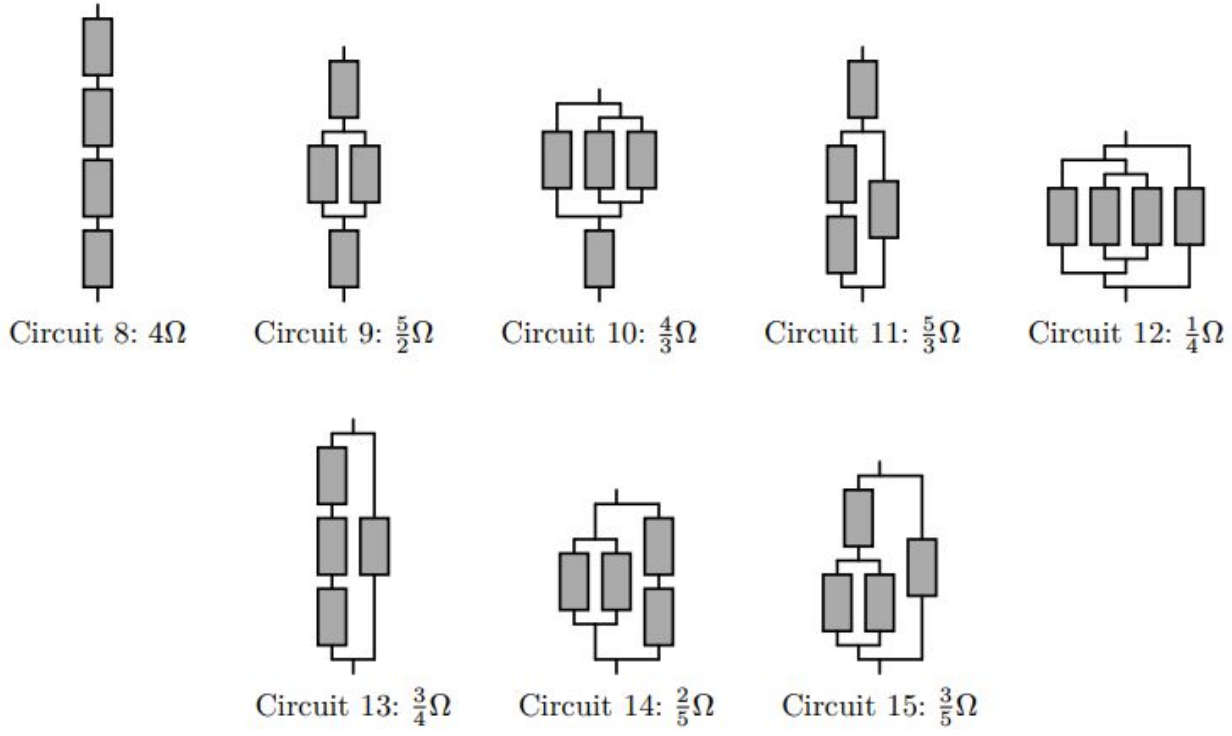


The Definitive Guide to Resistor Networks

By “metamuffin”, 2026.

https://s.metamuffin.org/projects/resistor-networks.pdf

“I generated all resistor networks in rust (takes about 50s on one core), then I load them in typst for rendering and generating the actual book (takes about 40s); this also required a custom typst build with increased recursion limits :)”



“Is the List of Incomplete Open Cubes Complete?”

Natasha Rozhkovskaya, Kansas State University, 2014

https://www.math.ksu.edu/%7Erozhkovs/LeWitt_cubes.pdf

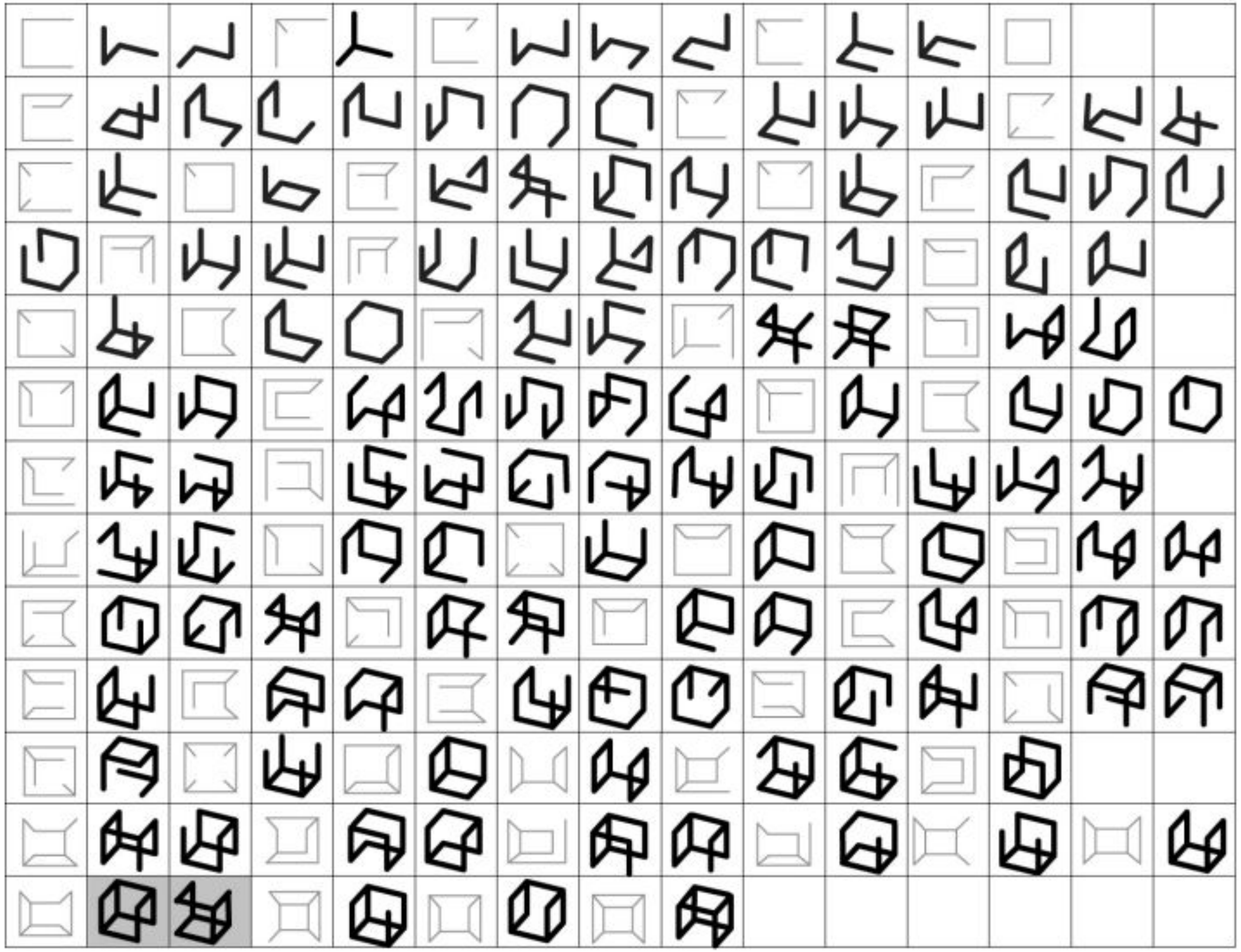


Figure 14: Embeddings of cubical graphs in I^3.

SmallCategories

Ben Spitz, https://smallcats.info/
("in beta")

A "category" is a mathematical object consisting of dots with arrows in between them, obeying a couple simple rules.

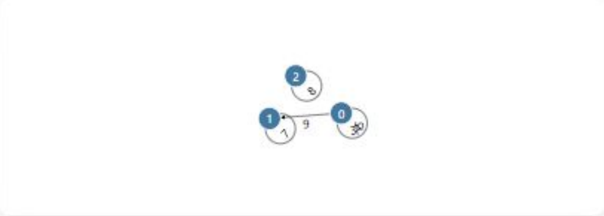
164,975 categories each with properties listed, the "multiplication table" of composition, and a visualization.

Quick Reference

Web ID	6ca55ed1-f425-4004-a199-29def9560838
Morphisms	9
Objects	3
Index	59248
Name	N/A
Description	N/A

Visualization

This feature is still janky! Please excuse the mess.



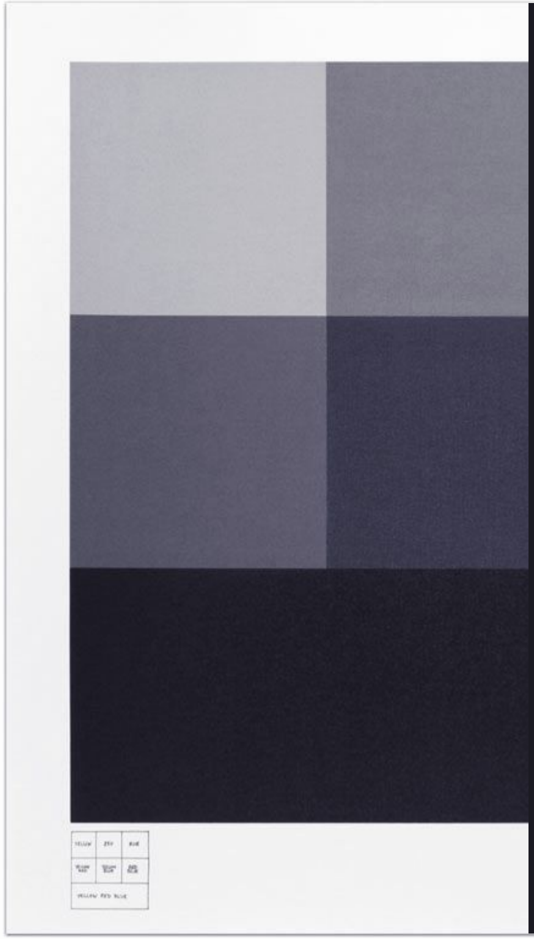
Morphisms 0 through 2 are identities, shown as objects in this visualization.

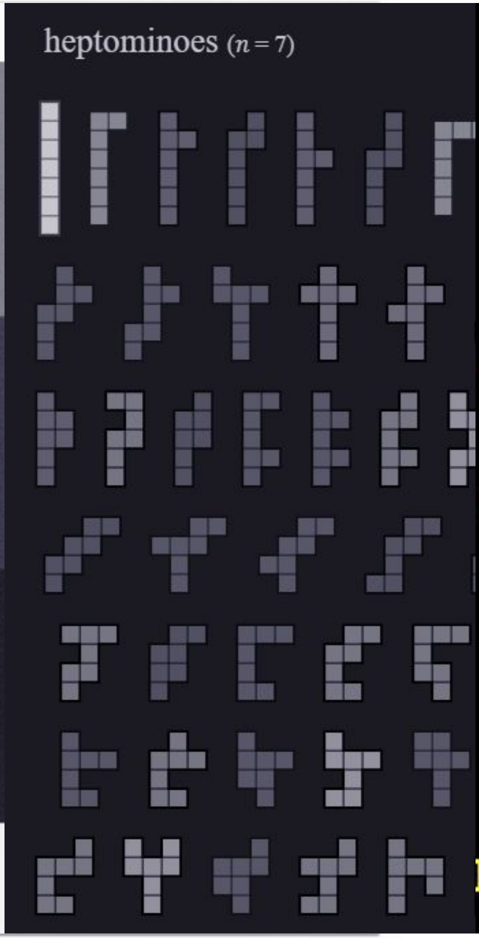
Table

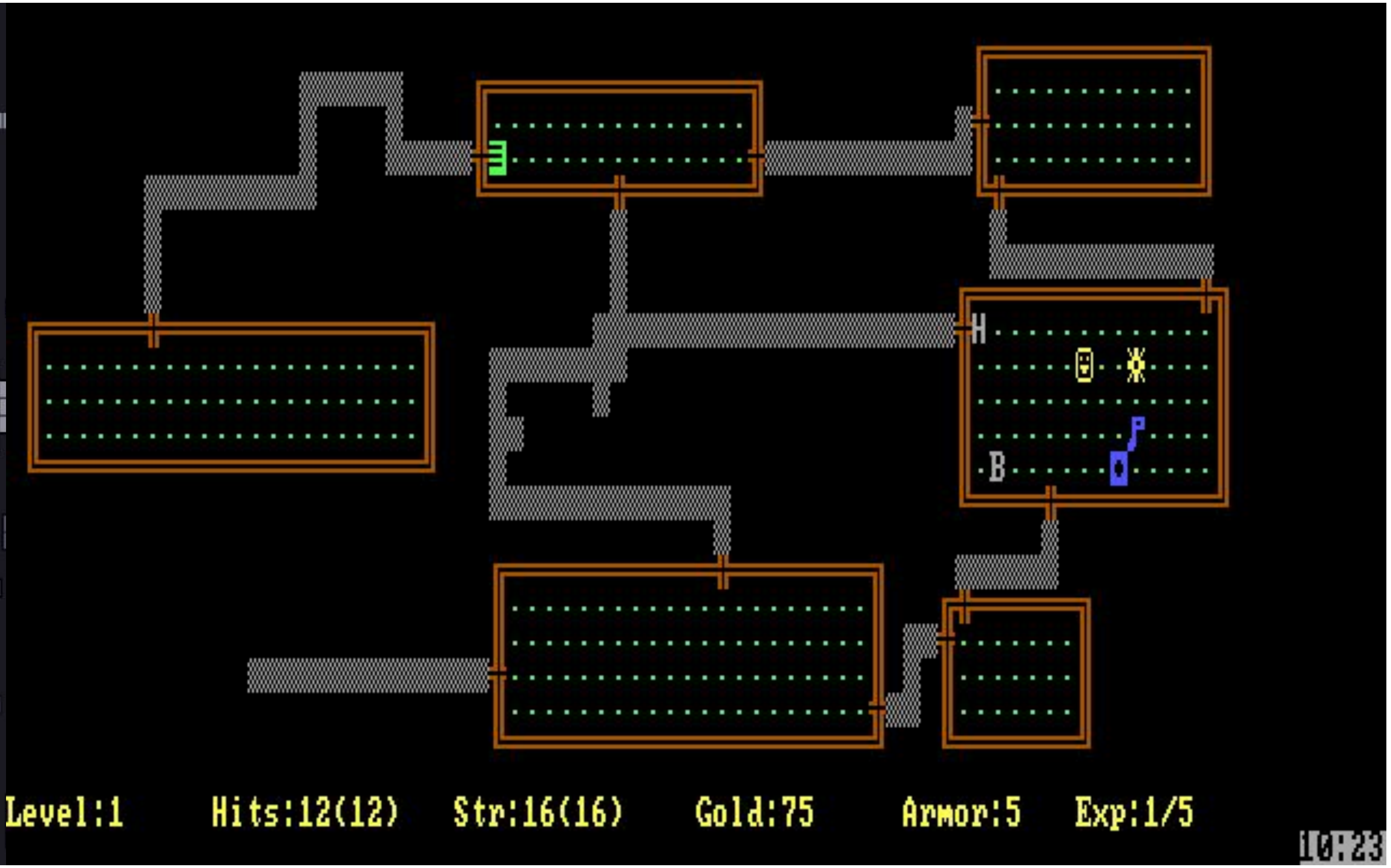
row ∘ col	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
0	0	/	/	3	4	5	/	/	/
1	/	1	/	/	/	/	6	7	/
2	/	/	2	/	/	/	/	/	8
3	3	/	/	3	4	5	/	/	/
4	4	/	/	3	4	5	/	/	/
5	5	/	/	3	4	5	/	/	/
6	6	/	/	6	6	6	/	/	/
7	/	7	/	/	/	/	6	7	/
8	/	/	8	/	/	/	/	/	2

Facts

✕ has_terminal_object ✕ has_binary_products
✕ is_discrete ✕ is_terminal ✕ has_initial_object
✕ has_binary_coproducts ✕ is_groupoid ✕ is_monoid
✕ is_connected ✕ has_finite_products ✕ is_preorder
✓ is_skeletal ✕ has_equalizers ✕ has_finite_coproducts
✕ is_complete ✕ is_cocomplete ✕ has_coequalizers
✕ is_initial




heptominoes ($n = 7$)


H
B
Level:1 Hits:12(12) Str:16(16) Gold:75 Armor:5 Exp:1/5
10:23

… and Rogue

Rogue

Developed by Michael Toy, Glenn Wichman, Ken Arnold, and Jon Lane starting around 1980.

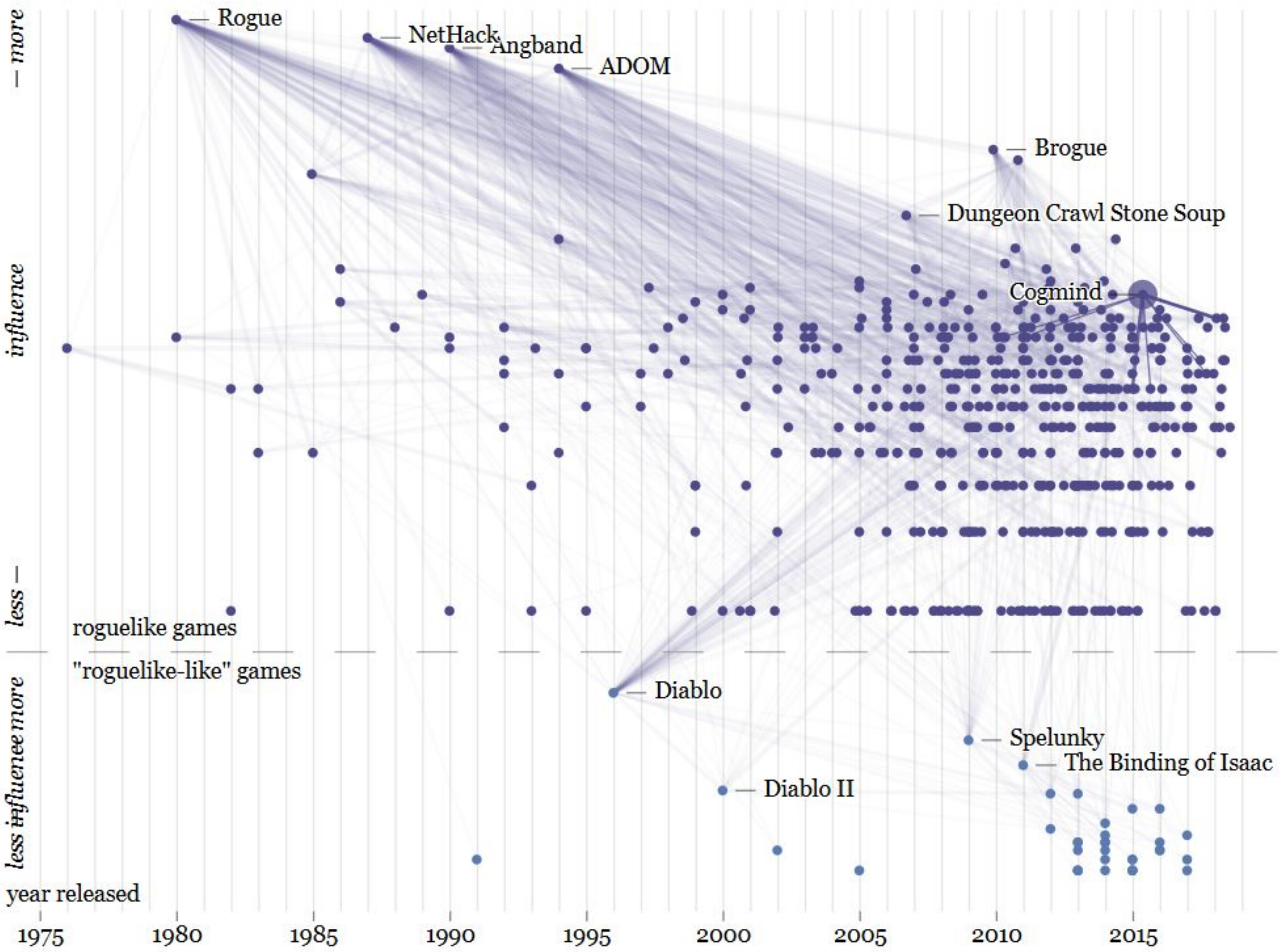
- One of the first Dungeons & Dragons-inspired computer games (though those started showing up *immediately* upon its release)
- Uses the curses library (by Ken Arnold) for drawing text-based interfaces
- Spread along with Unix as part of the Bekreley Software Distribution

A genre-definer for “Roguelikes”

- Procedurally generated
- No save points (permadeath), Interacting systems, resource management, grid
- Originally mainly text-based

Cloned frequently in the 80’s, 90’s, and still an influence today.

Influence graph from https://spaxe.github.io/roguelike-universe/



Rogue Level Generator

9 rooms

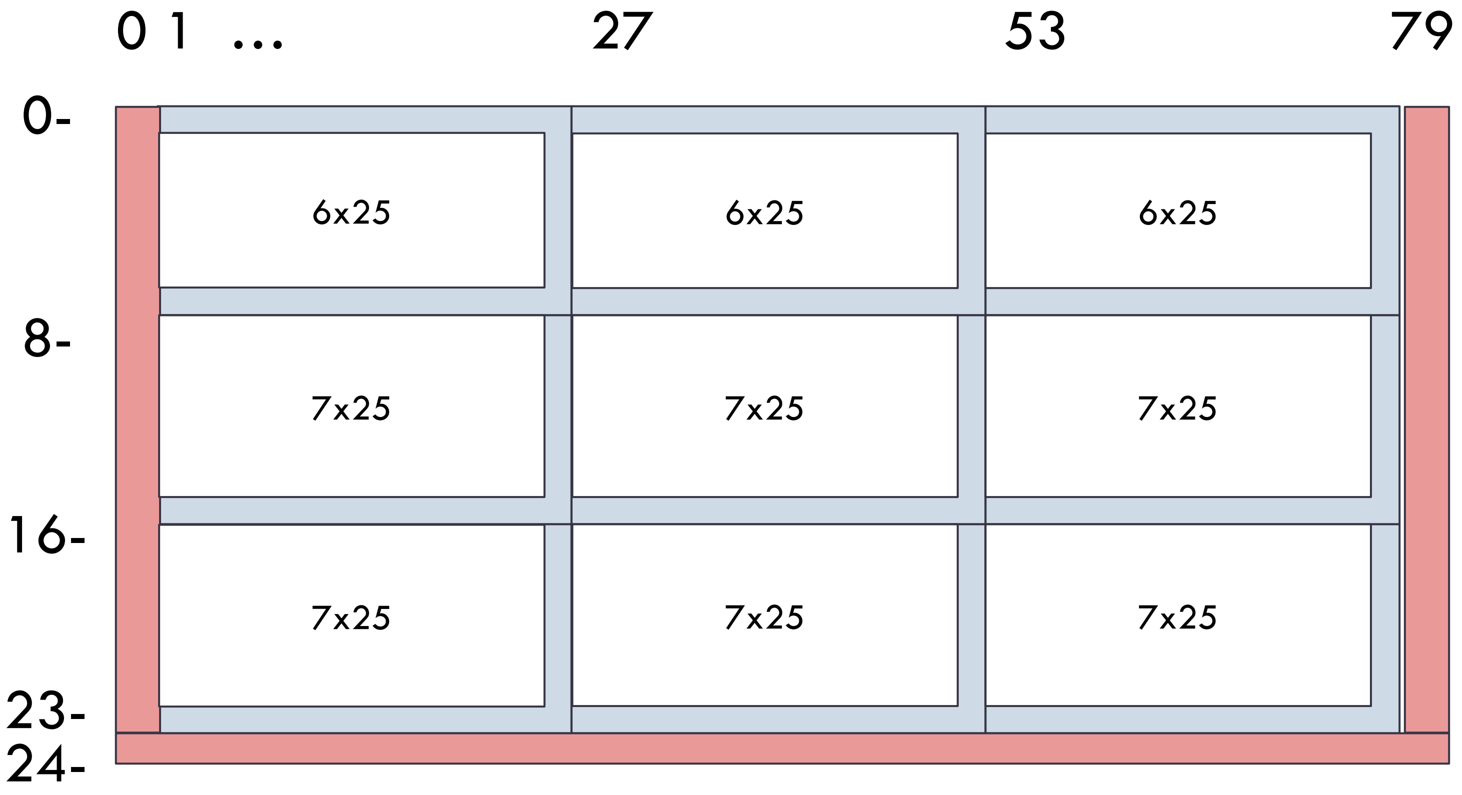
Up to 3 can be missing

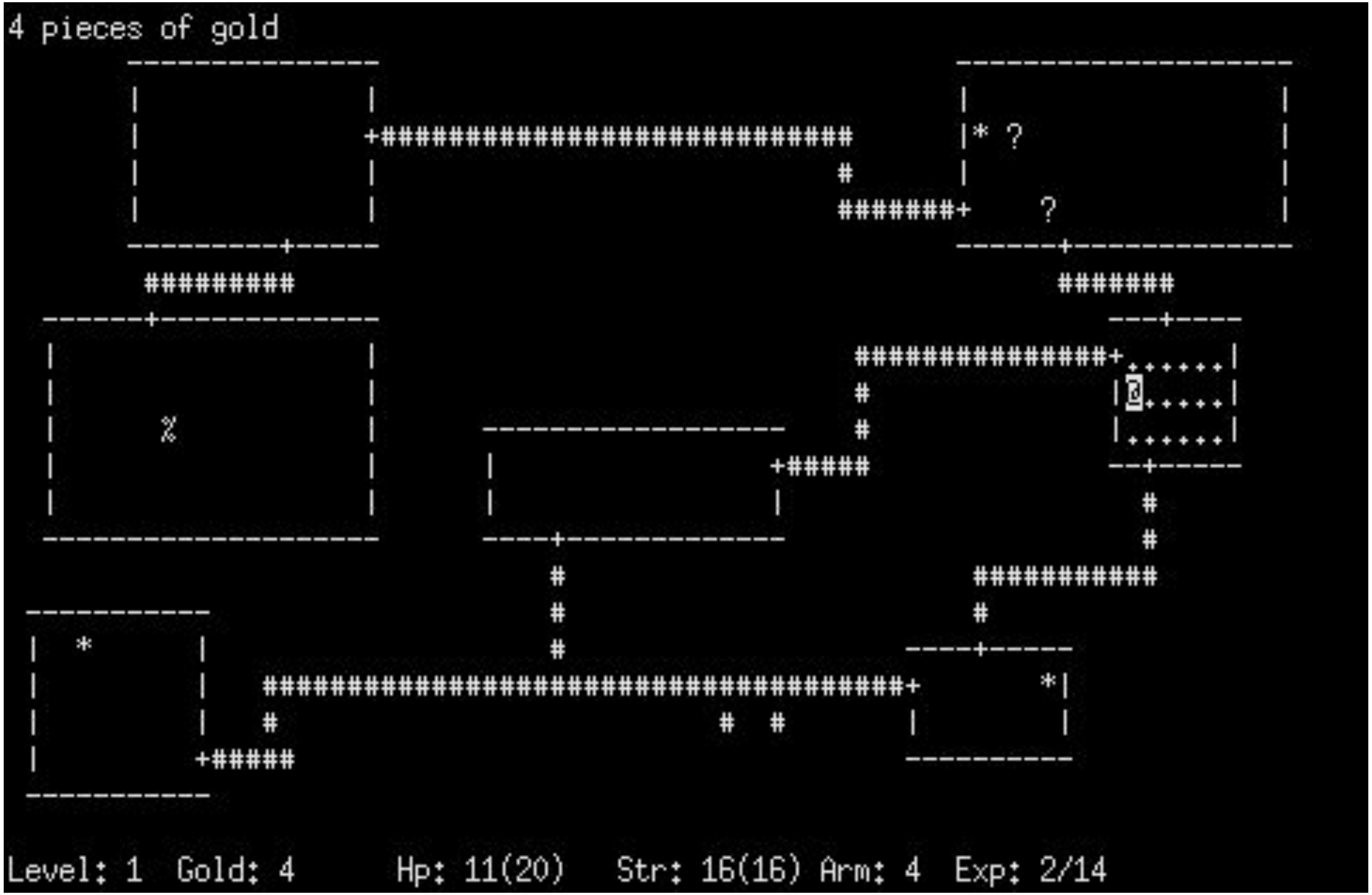
For each room:

- Pick a width [4,25]
- Pick a height [4,7]

Pick a location within the “cell” that fits.

Start over if the room would be on the first row (so top rooms only have height 4-6.)





https://commons.wikimedia.org/wiki/File:Rogue_Screenshot.png

Counting the Possibilities

Width 25	Width 24	….	Width 5	Width 4	
1 horizontal position	2 horizontal positions	…	21 horizontal positions	22 horizontal positions	=253 possibilities

Same calculation for height gives 6 (top row) or 10 (other two rows) possibilities.

So, there are about (253 x 6)^3 x (253 x 10)^6 possible combinations of room sizes, if no rooms are absent. That's about 2^(99.5) possibilities.

- P.S. this number doesn't change much if we account for "gone" rooms

Rogue as Serial Art

Rogue uses a PRNG (a linear congruential generator) with a 32-bit seed value.

There are only 2^32 possible versions of a level!

We could enumerate all of them.

- As an art project?
- To better understand the generator?
- I just thought it was cool that not only are there room configurations that are impossible, they *vastly* outnumber the ones that are valid.

The Rogue RNG(s)

Original rogue:

```
(((seed = seed*11109+13849) >> 16) & 0xffff)
```

This is a standard Linear Congruential generator, but using bits 16-32 of the seed avoids some of its worst behaviors.

MS-DOS port:

```
seed *= 125;
seed -= (seed/2796203) * 2796203;
return ((ran() + ran())&0x7fffffffl)
```

“This is adapted from the FORTRAN version in ‘Software Manual for the Elementary Functions’ by W.J. Cody, Jr and William Waite.”

Known features of LCGs

- **Marsaglia's theorem:** if used to choose points in an *n*-dimensional space, the points will lie on at most (*n*! * 2^32)^(1/*n*) hyperplanes
  - e.g., for n=4 that is 567
- **No duplicate values** until full cycle completed.
- **Low-order bits have short period;** for example the lowest bit alternates between 0 and 1 each time.

Taking high bits mitigates most of these failures.

Other potential problems

If we compute rng() mod K, and K is not a power of 2, then our distribution will be nonuniform:

- 65,536 possible values of the upper 16 bits
- K=22 for room width
- Values 0 through 19 have probability 2979/65536
- Values 20 through 21 have probability 2978/65536

This is probably too small to have any discernible effect.

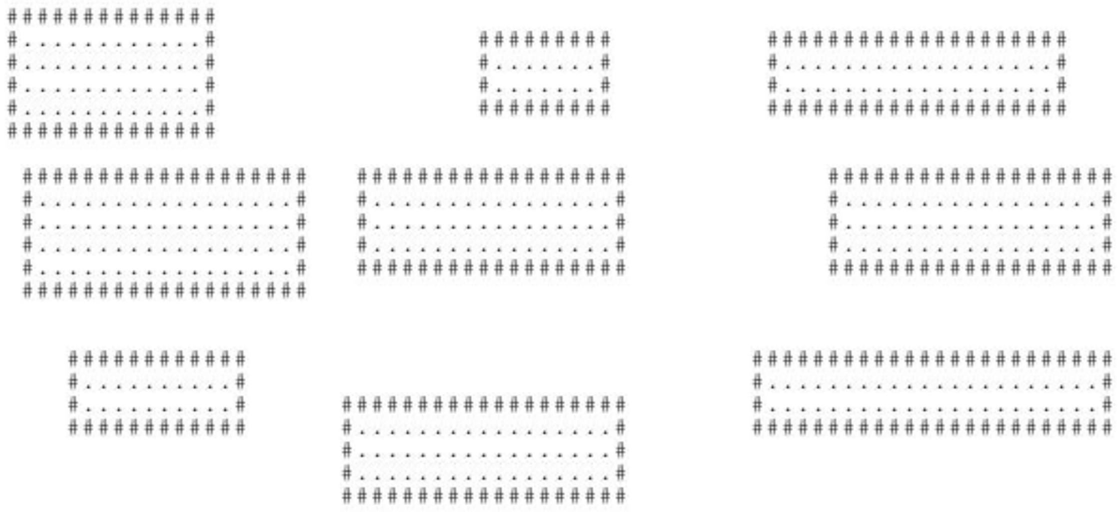
It's not even hard!

I thought that this would take some nontrivial engineering work.

- Parallelize, implement on a GPU, ...?
- Implement an index for reverse lookup ...?

Nope, I copied the implementation into a C++ program (*) and it only takes 11 minutes to run through every possible seed.

(*) = turns out it is not possible to copy Rogue's room generation faithfully without also copying its monster and item generation, oops



first 100 seeds

S3 storage – about $5/month

Amazon S3 > Buckets > rogue-star-bitmaps

Objects (999+)

Copy S3 URI | Copy URL | Download | Open | Delete | Actions | Create folder

Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them perm
more

Find objects by prefix

< 1 2 3 4 5

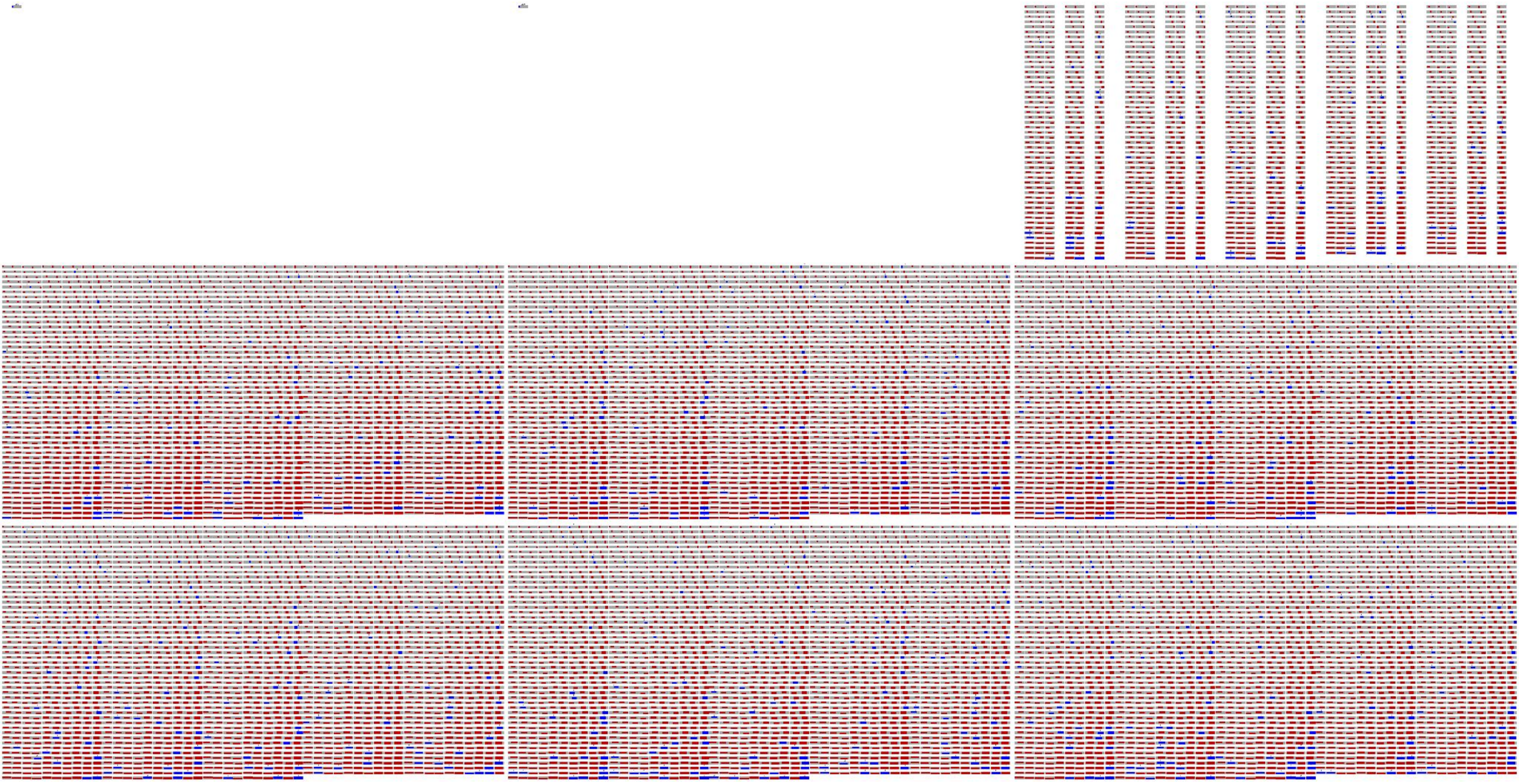
Name	Type	Last modified	Size	Storage class
00000000-room0-height4	-	July 12, 2025, 15:15:39 (UTC-05:00)	4.0 MB	Standard
00000000-room0-height5	-	July 12, 2025, 15:15:39 (UTC-05:00)	4.0 MB	Standard
00000000-room0-height6	-	July 12, 2025, 15:15:39 (UTC-05:00)	4.0 MB	Standard
00000000-room0-width10	-	July 12, 2025, 15:15:39 (UTC-05:00)	4.0 MB	Standard
00000000-room0-width11	-	July 12, 2025, 15:15:39 (UTC-05:00)	4.0 MB	Standard
00000000-room0-width12	-	July 12, 2025, 15:15:39 (UTC-05:00)	4.0 MB	Standard
00000000-room0-width13	-	July 12, 2025, 15:15:39 (UTC-05:00)	4.0 MB	Standard
00000000-room0-width14	-	July 12, 2025, 15:15:39 (UTC-05:00)	4.0 MB	Standard
00000000-room0-width15	-	July 12, 2025, 15:15:39 (UTC-05:00)	4.0 MB	Standard
00000000-room0-width16	-	July 12, 2025, 15:15:39 (UTC-05:00)	4.0 MB	Standard
00000000-room0-width17	-	July 12, 2025, 15:15:41 (UTC-05:00)	4.0 MB	Standard

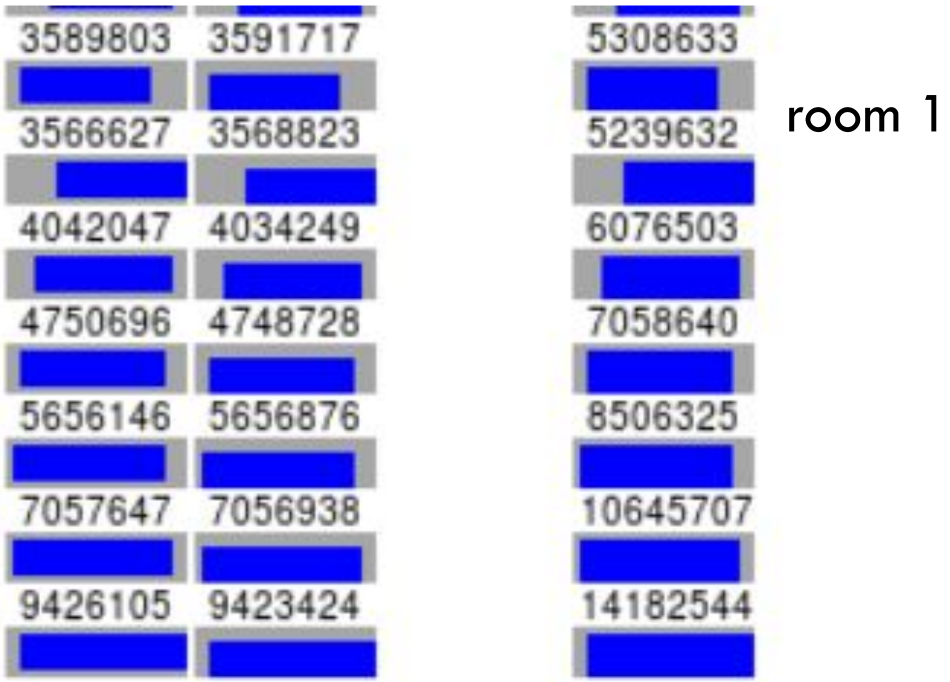
Postselection

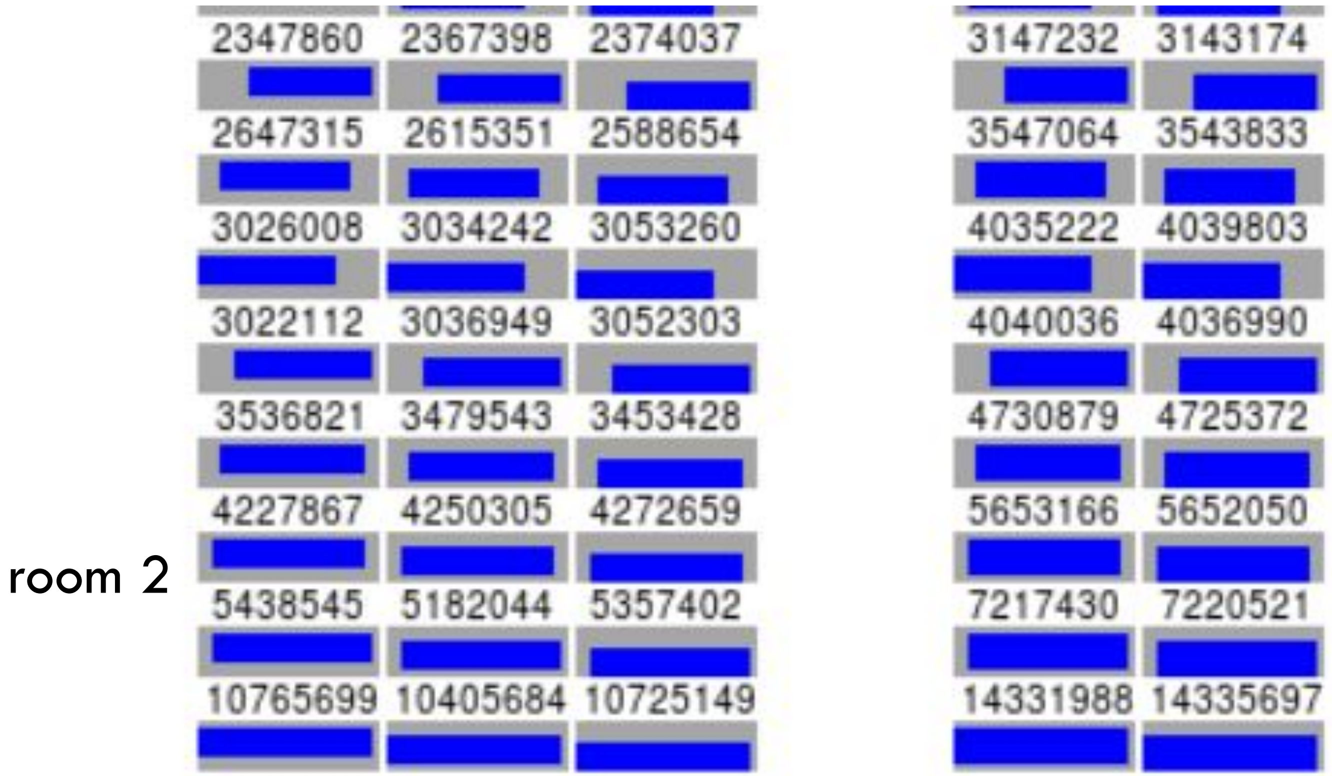
What if the top-left room is 4x4?

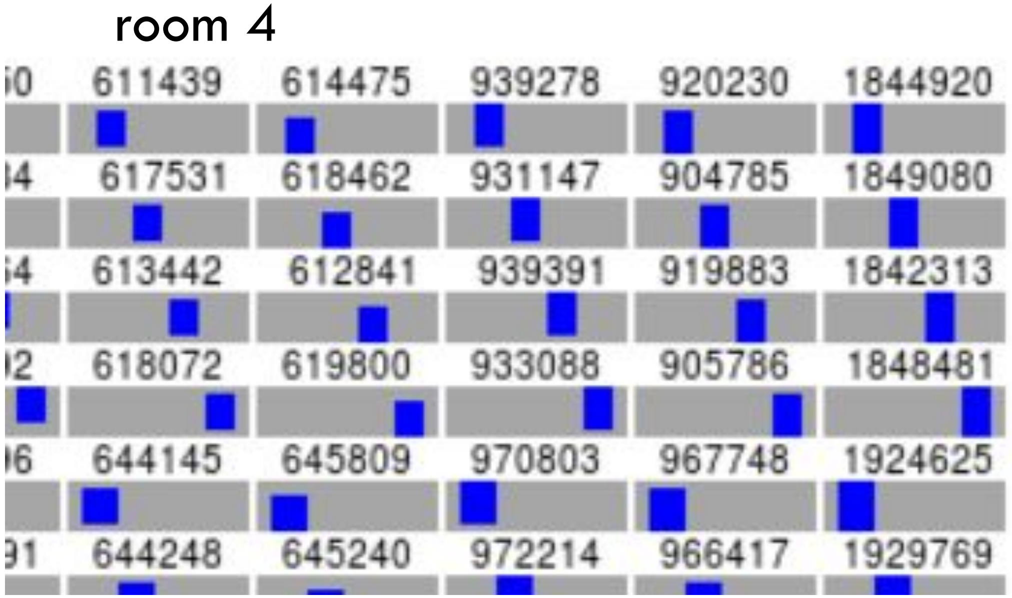
This happens 63,579,000 times. The distribution of the remaining rooms is about even.

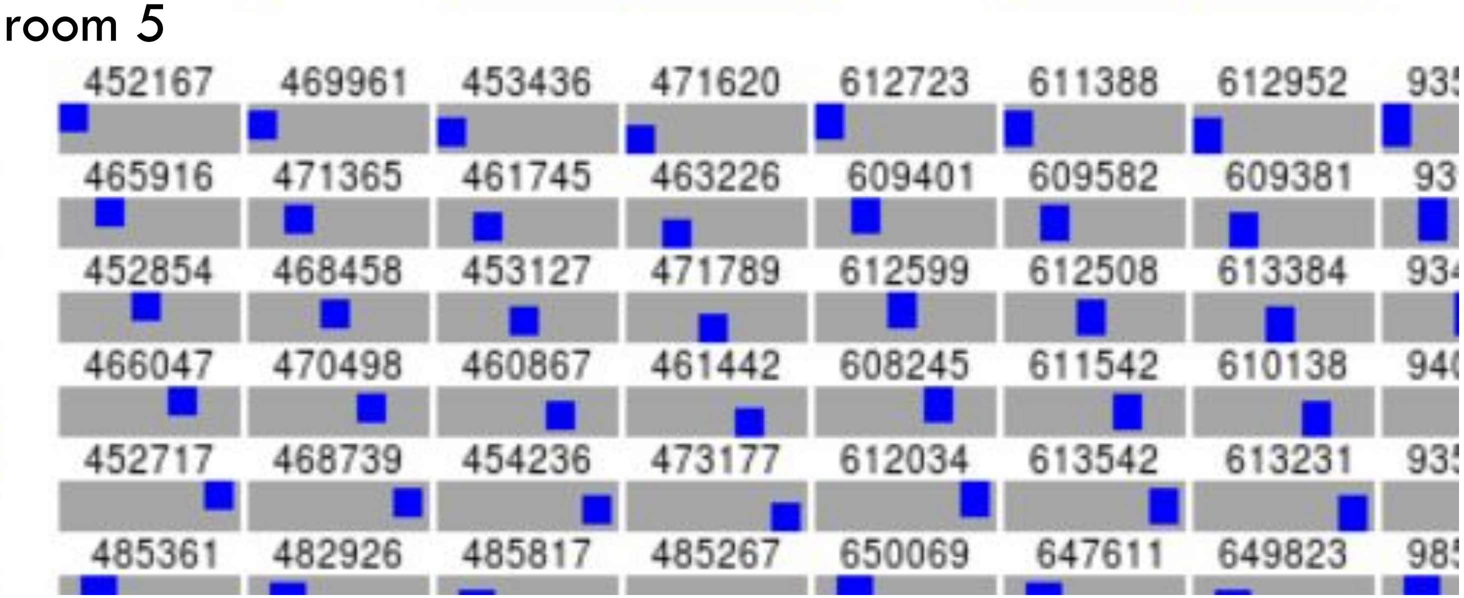
(This plot shows how what % of time the given square is occupied.)




room 1
3589803 3591717 5308633
3566627 3568823 5239632
4042047 4034249 6076503
4750696 4748728 7058640
5656146 5656876 8506325
7057647 7056938 10645707
9426105 9423424 14182544


room 2
2347860 2367398 2374037 3147232 3143174
2647315 2615351 2588654 3547064 3543833
3026008 3034242 3053260 4035222 4039803
3022112 3036949 3052303 4040036 4036990
3536821 3479543 3453428 4730879 4725372
4227867 4250305 4272659 5653166 5652050
5438545 5182044 5357402 7217430 7220521
10765699 10405684 10725149 14331988 14335697


room 4
611439 614475 939278 920230 1844920
617531 618462 931147 904785 1849080
613442 612841 939391 919883 1842313
618072 619800 933088 905786 1848481
644145 645809 970803 967748 1924625
644248 645240 972214 966417 1929769

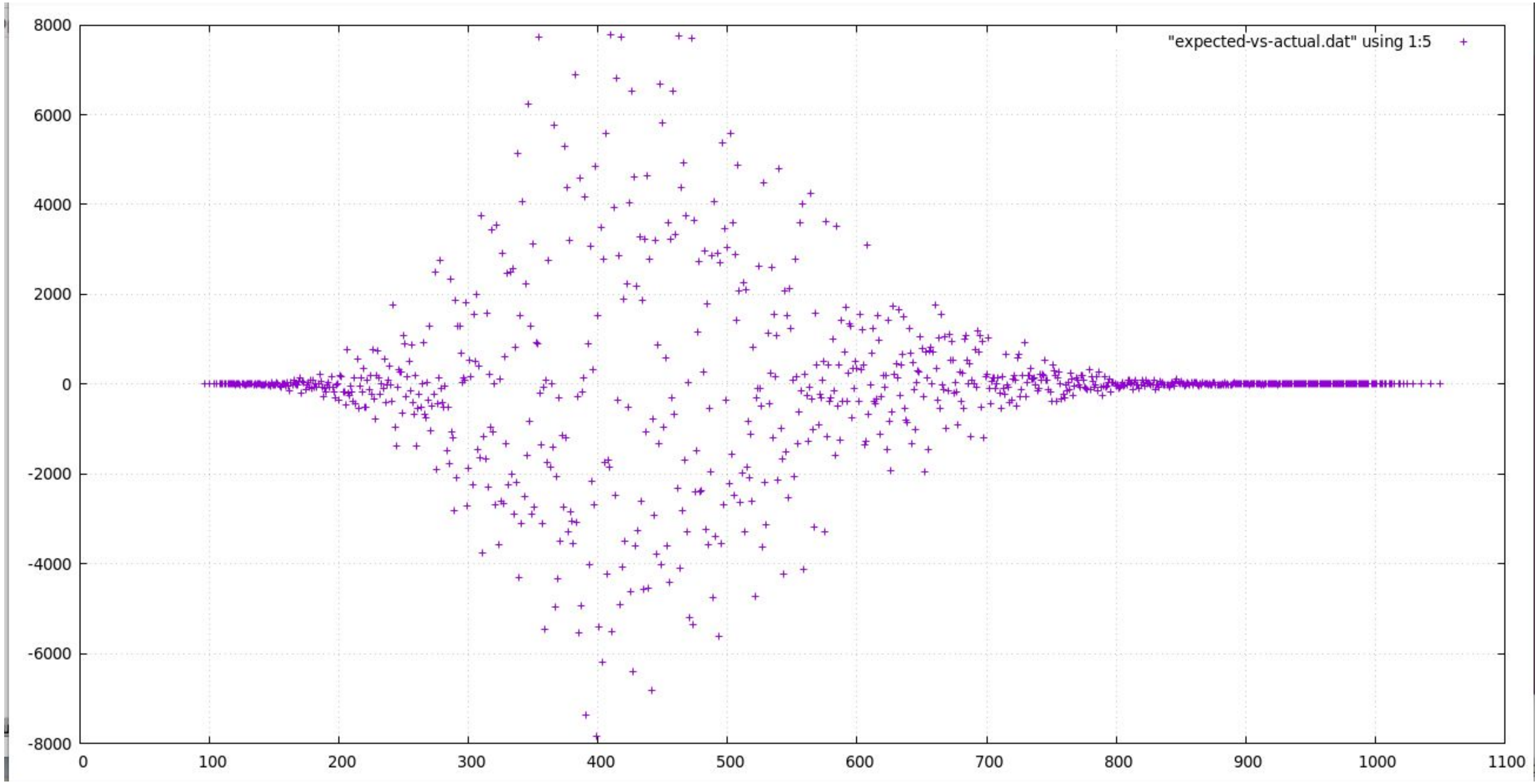

room 5
452167 469961 453436 471620 612723 611388 612952
465916 471365 461745 463226 609401 609582 609381
452854 468458 453127 471789 612599 612508 613384
466047 470498 460867 461442 608245 611542 610138
452717 468739 454236 473177 612034 613542 613231
485361 482926 485817 485267 650069 647611 649823

Finding extremal seeds

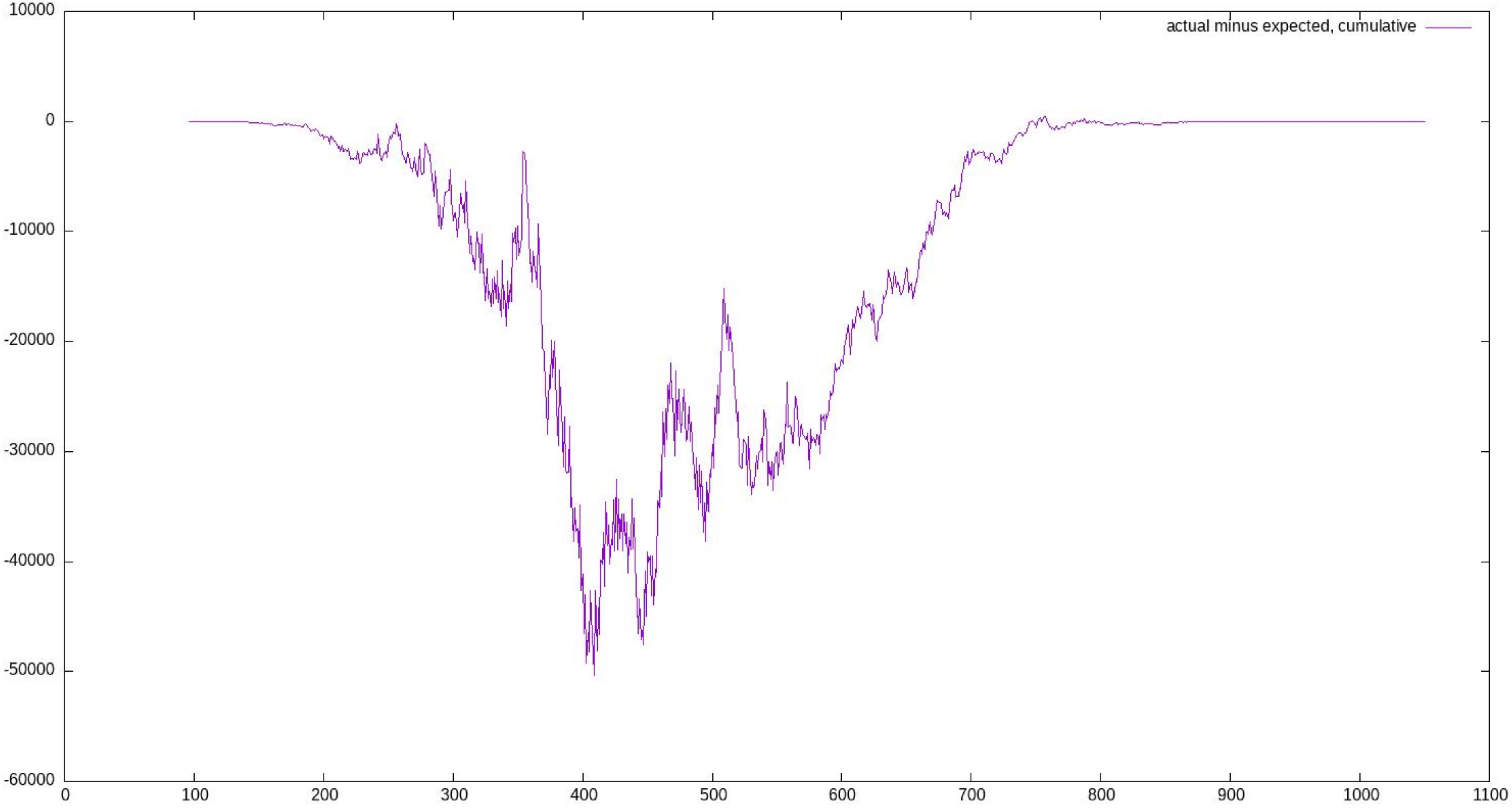
total room area	actual / cumulative	expected cumulative
96-110	0	2
112	2	7
113	4 (6)	9
116	7 (13)	21
117	6 (19)	26
118	5 (24)	30
120	18 (42)	55
121	16 (58)	69
122	11 (69)	82

There seems to be a deficit in how often the smallest areas appear, compared to their expected distribution (in the case where three rooms are gone, so 25% of the sample space.)

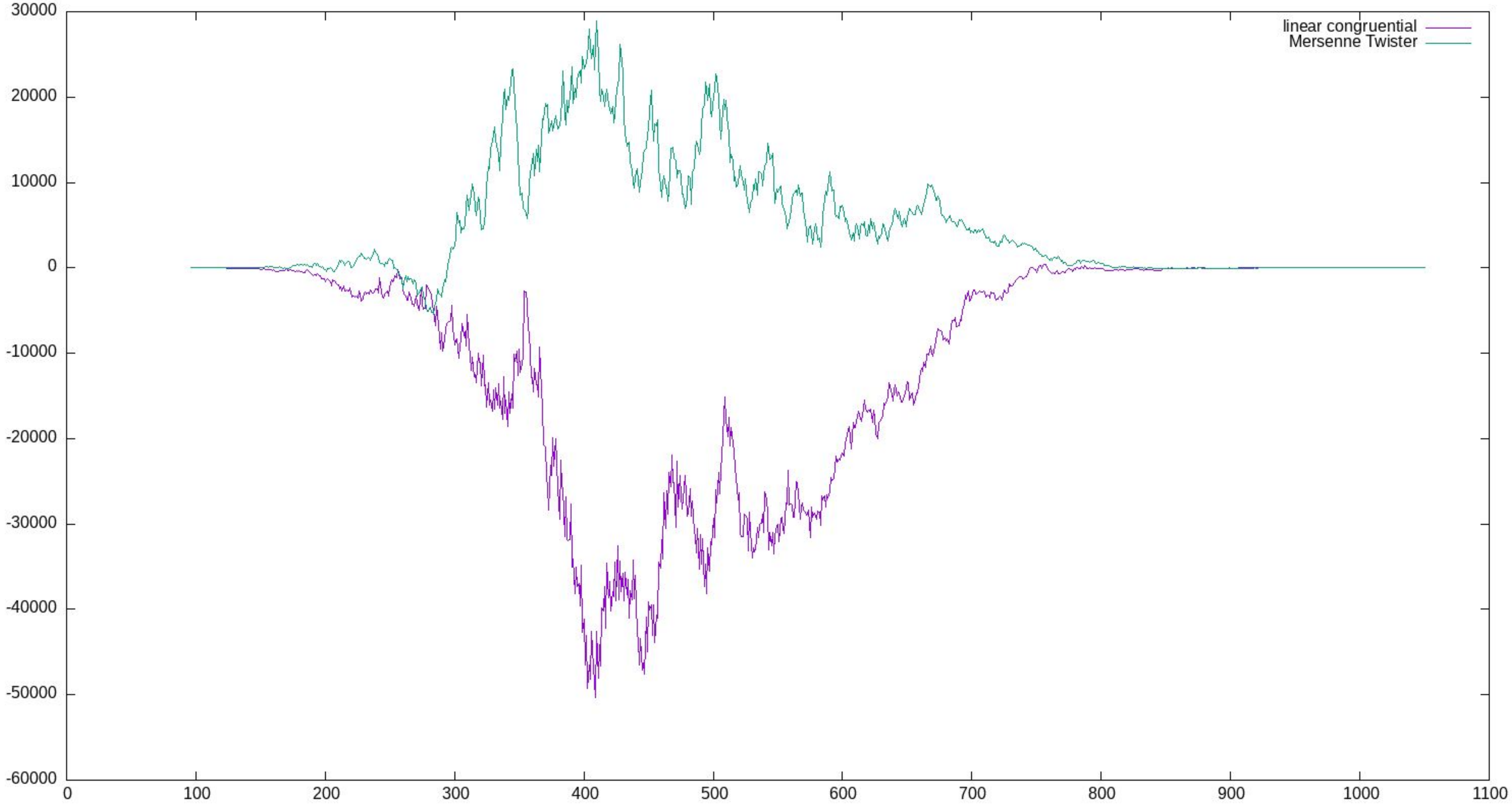
Expected vs. actual (per size)



Expected vs. actual (cumulative)



Expected vs actual (comparison)



What I’d like to find (but haven’t)

- Some artifact of the room distribution that I can point to and say “this is because Rogue uses a linear congruential generator.”
- Some better visualization of the 36-dimensional space of room configurations

Things to think about

Generators that use 32-bit seeds can be *fully* explored.

- As science!
- As serial art!
- As CYA?!?
- To completely solve a nontrivial roguelike game?

What are other connections between Serial Art and procedural generation?

- I'd love to hear an actual art historian or critic weigh in!

Are there natural examples of games with seeds in the hundreds or millions rather than in the billions?

Things to think about (2)

Building a solver for real roguelike games– could easily "reconstruct" the seed by mistake. :(

- Once we have seen as little as 3 rooms, the value of the initial random seed can be determined, which turns the game from a random one into a deterministic one.

Dan Lew: the most erratic Balatro Decks

Dan Lew is speaking on Passkeys at Minnebar this year, but here's some enumerative work from him on another genre-defining game, Balatro:

https://blog.danlew.net/2025/11/03/balatros-least-erratic-decks/

- Balatro rolled its own PRNG (why? who knows?), with 2.3 trillion possible seeds
- The "erratic" deck randomizes all the cards in your starting deck.

"**I3EFEF17** has the highest number of the same rank w/ 25 tens, followed by **RGJO14OZ** w/ 24 fours.

8778L6US has the highest number of the same suit w/ 39 hearts.

77XX2TEK has the best combination of same suits & ranks, with 20 fours and 38 spades."

THANK YOU!

On Mastodon: `@markgritter@mathstodon.xyz`
On Bluesky: `@markgritter.bsky.social`
On Github: `mgritter/rogue-room-generation`
In Real Life: Principal Engineer at thirdlaw.io



More Sol LeWitt! at NYMOMA and SFMOMA

PICO-8 RNGs

How many seeds are there in PICO-8? Looks like 32 bits, though the state is 64 bits.

Strangeness in the PICO-8 RNG: https://www.lexaloffle.com/bbs/?pid=81103

```
function rng.rnd(n)
  n=n or 1
  rng.hi=(rng.hi<<0x10|rng.hi>>>0x10)+rng.lo
  rng.lo+=rng.hi
  return ((rng.hi<0 and 0x8000-n or 0)%n+(rng.hi&0x7fff.ffff)%n)%n
 end
```

This lead to a fix! Not sure what the current PRNG is.

Pico-8

Random movement in the Pico-8 virtual machine leads to stairsteps! Not random at all.